

From Open Weights to Decision Policies: A Controlled Study of Scale, Conversion Method and Task Structure in Language-Model Decision-Making

DeadEye project

Draft built 8 October 2026

Abstract

Small open-weight language models are attractive as decision-makers because they are cheap and fast. Yet the same weights can be prompted, scored, probed or fine-tuned into a policy, and how this choice interacts with scale has not been measured under controlled conditions. We introduce DEAD EYE, a benchmark and open harness for this measurement. It has seven seeded decision tasks with known oracles (a multi-armed bandit, a contextual bandit, loan approval under distribution shift, gridworld navigation, tic-tac-toe, blackjack and customer-support triage under a written policy) and six conversion methods: the assistant’s own free reply as the unconverted reference, free-form generation, exact likelihood scoring of legal actions, a linear probe on hidden states, LoRA behavior cloning, and a probe on raw task features that uses no language model. Returns are normalized so that a random policy scores 0 and the oracle 1, and common random numbers pair every comparison. We test seven pre-registered hypotheses on within-family ladders from 70M to 72B parameters. Under generation, normalized score changes by [result pending: [slope $\hat{b}$, 95% CI]] per tenfold increase in parameters. A [result pending: [X]]B model fine-tuned with LoRA matches a [result pending: [Y]] $\times$ larger zero-shot model on [result pending: [k]] of six tasks, [result pending: [but not / and also]] under shift, and likelihood scoring closes [result pending: [Z]]% of the gap between the smallest and largest models. Code, configurations and per-decision logs are public.

Draft. This is the complete draft of the paper with the empirical results still to be measured: every blue “result pending” mark is a number the main sweep will fill in, and every red TODO is an open item. Figures and tables that read “placeholder” are generated from the results by the report tool.

1 Introduction

Language models increasingly act rather than answer. They choose moves in games, calls to tools and steps in workflows, and what they produce is executed instead of read (e.g. [Yao et al., 2023](#); [Liu et al., 2024](#); [Paglieri et al., 2025](#)). Many of these systems run on open weights, and small open-weight models are attractive for them because they are cheaper, faster and can run locally. Anyone building such a system faces a practical question: how small can the model be before its decisions get worse, and what can be done about it when they do?

The literature gives mixed signals. Pre-training loss falls predictably as parameters and training data grow ([Hoffmann et al., 2022](#)), but decision quality need not follow. Some abilities appear abruptly at scale ([Wei, Tay, et al., 2022](#)), although many such jumps turn out to be artifacts of discontinuous metrics ([Schaeffer, Miranda and Koyejo, 2023](#)). Several failures persist as models grow. Language models explore poorly in multi-armed bandits unless the history is summarized for them ([Krishnamurthy et al., 2024](#)). Models from 2B to 27B parameters leave much of the action space untried and often act against their own correct rationale ([Schmied et al., 2026](#)), and they can fail even to exploit what they have already seen ([Harris and Slivkins, 2026](#)). Other results favor small models. Given extra test-time compute, a small model can beat one 14 times its size in FLOPs-matched comparisons, on problems it already solves some of the time ([Snell et al., 2025](#)), and distilling bandit algorithms into smaller models let them explore better than larger models without that help ([Nie et al., 2025](#)). Most agent benchmarks evaluate frontier APIs or heterogeneous collections of open models (e.g. [Wu et al., 2024](#); [Liu et al., 2024](#); [Duan et al., 2024](#); [Paglieri et al., 2025](#); [Ruoss et al., 2025](#)), so when one model beats another they cannot tell whether size or training recipe made the difference.

There is a second, less discussed variable. A language model is a distribution over text, not a policy, and something has to turn one into the other. The usual choice is to generate text and parse an action out of it, which mixes formatting failures with bad decisions (Huang *et al.*, 2022; Liu *et al.*, 2024). One can instead score every legal action by its likelihood (Carta *et al.*, 2023; Tan *et al.*, 2024), read the decision off the hidden states with a probe (Belinkov, 2022; Orgad *et al.*, 2025), or fine-tune on expert actions (Chu *et al.*, 2025). These methods use the weights in different ways, and in multiple-choice question answering they disagree with each other more often than one would hope (Robinson and Wingate, 2023; Wang *et al.*, 2024). Papers on them usually study one method at one or two model sizes. A small model that fails when prompted might succeed when scored or probed; we do not know how often that happens, or whether it depends on the kind of decision.

The closest existing studies each cover part of this space. Jiwatode *et al.* (2026) vary Qwen3 model size, thinking mode and planning horizon on three spatial-navigation games, with one model family and prompting as the only conversion. Schmied *et al.* (2026) study models from 2B to 27B on bandits, contextual bandits and tic-tac-toe, identify greediness, frequency bias and a knowing-doing gap, and show that RL fine-tuning on the model’s own reasoning increases exploration and narrows the knowing-doing gap; their comparison is between prompting and RL fine-tuning, not across a set of conversion methods on a size ladder. Monea *et al.* (2025) span 500M to 70B parameters, but for in-context bandit learning on classification tasks, and Nie *et al.* (2025) relate bandit regret to model size and test inference-time and distillation interventions. None of these crosses same-recipe model ladders with several conversion methods and several kinds of decision problem, normalizes against an oracle, or fixes its hypotheses before collecting data. Some of our hypotheses already have partial answers in this work, and Section 2 says where. For those, what we add is a controlled, within-family and pre-registered measurement, not a new direction of effect.

Our contributions are the following.

1.0.0.1 Research questions. We ask five questions, the first four pre-registered and the fifth exploratory. RQ1: within a model family trained with one recipe, how does decision quality scale with parameter count, and does the slope depend on the structure of the task? RQ2: does the conversion method (free reply, formatted generation, likelihood scoring, a probe on hidden states, LoRA behavior cloning) change that dependence on scale? RQ3: which other factors move decision quality, and by how much: instruction tuning, quantization, reasoning budget, observation format, prompt encoding, and the data budget of adaptation? RQ4: what is the frontier of decision quality against inference cost, measured as latency, tokens and parameters? RQ5: how do small open-weight models converted with transparent methods compare with the purpose-built decision models that appeared in late 2026, on the same tasks, seeds and metrics, including the calibration of the probabilities they return?

- **A benchmark.** Seven procedurally generated decision tasks, each with an exact oracle, covering exploration (a multi-armed bandit), in-context rule learning (a contextual bandit), one-shot tabular decisions under covariate shift and under a reversed effect (loan approval), deterministic planning (gridworld), adversarial play (tic-tac-toe), risk (blackjack) and the application of a written policy to a customer’s request (support triage). A seed fixes an instance and all of its randomness, so policies are compared on identical potential outcomes, and returns are mapped onto a scale on which the random policy scores 0 and the oracle 1 (Section 3).
- **A harness.** Six methods behind one interface: the assistant’s unconverted free reply as the reference point, free-form generation, exact likelihood scoring of legal actions, a logistic probe on hidden states, LoRA behavior cloning, and the same probe applied to raw task features as a control that uses no language model. The harness runs local Hugging Face models and models behind any OpenAI-compatible server, logs every decision with the raw model output, and regenerates every result table and figure in this paper from those logs.
- **A pre-registered study.** Within-family ladders (Qwen2.5 from 0.5B to 72B, Qwen3, SmolLM2 and Gemma 3, with Pythia and OLMo 2 as fully open controls), paired seeds, and seven hypotheses whose falsification criteria are frozen at a git tag before the main sweep (Section 4).
- **Findings.** [result pending: [One sentence per hypothesis, for example: the slope under generation is b per tenfold increase in parameters and is steeper on planning tasks (H1); a LoRA-tuned model of at most 1.7B matches a ten

times larger zero-shot model on k of six tasks in distribution but not under shift (H3); scoring closes $Z\%$ of the small-to-large gap (H4); 4-bit weights cost q below 2B (H5); no prompted model reaches UCB1 on the bandit (H6); chain-of-thought helps planning only from mB upward (H7)] (Sections 5 and 6).

Section 2 reviews related work and marks where it already answers parts of our hypotheses. Section 3 defines the tasks, the score and the conversion methods, and Section 4 the models, factors and statistics. Section 5 tests each hypothesis against its pre-registered criterion. Section 6 examines errors, cost and the language-model-free control, and Section 7 sets out what the design cannot tell us. The appendices give every prompt verbatim, the task generators, the full pre-registration, the compute budget, a reproducibility checklist and the verified model identifiers.

2 Related work

2.0.0.1 Language models as agents and decision-makers. SmartPlay (Wu *et al.*, 2024) probes nine capabilities with six games, one of them a two-armed bandit, and finds that models differ by capability rather than by a single agentic skill. BALROG (Paglieri *et al.*, 2025) grades difficulty from BabyAI to NetHack; models make progress on the easy environments and fail on the hardest. LMACT (Ruoss *et al.*, 2025) shows that even 512 expert episodes in context rarely bring frontier models to expert level on tasks that include tic-tac-toe and grid navigation. AgentBench (Liu *et al.*, 2024) evaluates 29 API and open models in eight environments and names weak long-term reasoning, decision-making and instruction following as the main obstacles, and GTBench (Duan *et al.*, 2024) finds that LLMs fail in complete-information deterministic games such as tic-tac-toe. Huang *et al.* (2022) showed that plans generated by language models are often not admissible actions, the problem that likelihood scoring avoids by construction. These benchmarks establish that difficulty depends on task structure, the premise of the task-dependent slope in H1, but their model sets are not controlled ladders and each fixes one conversion method. The design nearest to ours is that of Jiwatode *et al.* (2026), who cross Qwen3 model size with thinking mode and planning horizon on spatial games; we extend it to several families, five conversion methods and six decision structures. **2.0.0.2 Exploration and bandits.** Krishnamurthy *et al.* (2024) found that, across many prompt designs, only GPT-4 with chain-of-thought and an externally summarized history explored a multi-armed bandit reliably. EVOLvE (Nie *et al.*, 2025) relates regret to model size and shows that algorithm-guided inference and algorithm distillation let smaller models explore better than larger ones without such support. Schmied *et al.* (2026) trace the failures of 2B to 27B models to greediness, with up to 55% of the action space left unexplored, to a frequency bias that fades with scale, and to a knowing-doing gap: rationales were correct about 87% of the time, yet the chosen action often did not follow from them. Harris and Slivkins (2026) find that LLMs often fail even to exploit, falling short of simple linear regression, and Monea *et al.* (2025) show that models from 500M to 70B parameters can learn from reward in context. Algorithm distillation bears on how such policies should be trained: cloning the learning histories of a reinforcement-learning algorithm, not expert actions, is what teaches in-context reinforcement learning (Laskin *et al.*, 2023). We use UCB1 (Auer, Cesa-Bianchi and Fischer, 2002) as a realizable reference and standard regret definitions (Lattimore and Szepesvári, 2020). H6, that exploration is poor without scaffolding, is therefore largely established. What we add is its dependence on scale within families, its behavior under conversion methods other than prompting, and a separation of exploration errors from exploitation errors. **2.0.0.3 Scale, emergence and small models.** Pre-training loss falls predictably with parameters and training tokens (Hoffmann *et al.*, 2022), which motivates the log-linear form of H1. The same work complicates any reading of such a slope, because the small rungs of modern ladders are trained far beyond the compute-optimal number of tokens; SmoLM2-1.7B, for example, saw about 11T tokens (Ben Allal *et al.*, 2025). Parameter count therefore partly stands in for tokens per parameter. Emergent abilities (Wei, Tay, *et al.*, 2022) are the main alternative to a smooth slope, and Schaeffer, Miranda and Koyejo (2023) attribute many of them to discontinuous metrics, one reason we fit slopes to continuous normalized returns rather than to success rates. Few-shot ability improves steadily with scale within one recipe (Brown *et al.*, 2020), and the length of task a model can execute keeps growing with size after single-step accuracy has saturated (Sinha *et al.*, 2026). The case for small models rests on extra computation and adaptation. In FLOPs-matched comparisons a small model given more test-time compute can beat one 14 times larger on problems it already solves some of the time (Snell *et al.*, 2025), and algorithm distillation lets small models explore bandits better than larger unaided ones (Nie *et al.*, 2025), which partly answers the in-distribution half of H3 for that task family. **2.0.0.4 Converting a language model into a policy.** Scoring answer strings by probability is distorted by surface-form competition between paraphrases

(Holtzman *et al.*, 2021). Scoring an option symbol works only for models that bind symbols to options well (Robinson and Wingate, 2023), option identifiers carry a token-level prior (Zheng *et al.*, 2024), and few-shot predictions lean toward majority, recent and common labels (Zhao *et al.*, 2021). For instruction-tuned models, first-token probabilities often disagree with the answer the model writes (Wang *et al.*, 2024). Meaning-preserving changes of prompt format move few-shot accuracy by up to 76 points (Sclar *et al.*, 2024), and the serialization of tabular inputs matters for LLM classifiers (Hegselmann *et al.*, 2023). As a policy, likelihood scoring of legal actions underlies GLAM (Carta *et al.*, 2023) and TWOSOME (Tan *et al.*, 2024), and the latter shows that long action strings are penalized unless scores are normalized for length. This work motivates H4, but it was measured on question answering, not on sequential decisions across a size ladder.

A linear probe on frozen hidden states shows what is decodable, which is not the same as what the model uses, and needs control tasks to be interpreted (Belinkov, 2022). Intermediate layers are often better features than the last (Skean *et al.*, 2025), and models often encode the correct answer while generating a wrong one (Orgad *et al.*, 2025). That gap is how a probe could beat generation on our tasks, and it is why we pair the probe with a control trained on raw task features.

Behavior cloning (Pomerleau, 1988) compounds its errors once the learner leaves the expert’s state distribution (Ross, Gordon and Bagnell, 2011), and LoRA (Hu *et al.*, 2022) and QLoRA (Dettmers *et al.*, 2023) make it affordable at every rung of a ladder. Whether a cloned policy generalizes is contested. Supervised fine-tuning memorizes the rules it was trained on while outcome-reward RL generalizes (Chu *et al.*, 2025), which supports the out-of-distribution half of H3; frozen language models with RL-trained adapters (Szot *et al.*, 2024) and LoRA fine-tuning on human choice data (Binz *et al.*, 2025) generalize to some shifts, which cuts against it. LIMA (Zhou *et al.*, 2023) and URIAL (B. Y. Lin *et al.*, 2024) suggest that alignment mostly changes style and format. That predicts the second clause of H2, that base and instruct checkpoints converge under probing and fine-tuning, but it also threatens the first, since a base model with a good few-shot prompt may close the gap under prompting as well. DPO (Rafailov *et al.*, 2023), GRPO (Shao *et al.*, 2024) and self-play on text games (Liu *et al.*, 2026) are conversions that DEAD EYE v1 leaves out.

2.0.0.5 Reasoning and output format. Chain-of-thought helps large models, while small ones produce fluent but illogical chains (Wei, Wang, *et al.*, 2022). More reasoning can also hurt. Deliberation lowers accuracy on tasks where it hurts people too, such as implicit statistical learning (Liu, Geng, Wu, *et al.*, 2025); on controllable puzzles, thinking helps only at intermediate complexity (Shojaee *et al.*, 2025); longer reasoning can reduce accuracy (Gema *et al.*, 2025); and models of about 3B parameters or fewer do not reliably benefit from long chains (Li *et al.*, 2025). Strict output formats can suppress reasoning (Tam *et al.*, 2024), so our chain-of-thought prompt lets the model reason freely before a delimited action line. Every clause of H7 finds support somewhere in this work, but on different tasks and models.

Switching thinking on and off inside the same weights (Yang *et al.*, 2025) permits a cleaner test, and latency-normalized efficiency has not been compared across sizes and task structures. **2.0.0.6 Quantization.** At a fixed total bit budget, 4-bit weights are close to optimal on zero-shot language tasks (Dettmers and Zettlemoyer, 2023), and GPTQ (Frantar *et al.*, 2023) and AWQ (J. Lin *et al.*, 2024) quantize large models with little loss. Precision-aware scaling laws predict that the penalty for post-training quantization grows with the amount of pre-training data (Kumar *et al.*, 2025); small modern models are the most over-trained, so this is the mechanism behind H5. These studies measure loss, perplexity or accuracy on language benchmarks, so for H5 there is a mechanism but no direct evidence on multi-step decisions.

2.0.0.7 Evaluation methodology. We follow Agarwal *et al.* (2021) in reporting interquartile means and bootstrap confidence intervals, and Henderson *et al.* (2018) in fixing seeds and hyperparameter budgets across all rungs. Because results for language models depend heavily on prompt format and on the choice between scoring and generation (Sclar *et al.*, 2024; Wang *et al.*, 2024), we report both conversions side by side and release every decision. **2.0.0.8 Cognitive and behavioral studies of decisions.**

Studies in the style of experimental psychology ask how models decide, not only how well. GPT-3 outperformed people on a two-armed bandit but showed no directed exploration (Binz and Schulz, 2023). Even GPT-4 falls short of human-like rationality when it must form and keep beliefs (Fan *et al.*, 2024), language models show human-like risk and loss aversion (Jia *et al.*, 2024), and they assume that people are more rational than people are (Liu, Geng, Peterson, *et al.*, 2025). Our oracles are normative rather than human, but these findings suggest examining the direction of errors, for instance whether a model stands too early in blackjack, and not only their rate (Section 6).

2.0.0.9 Decision models. While this study was being prepared, “decision models” became a product category.

TypeSafe AI’s Jev, announced in September 2026, takes a state and typed questions and returns a chosen option with a

probability for every allowed answer and a confidence, never prose; the company describes it as a “System One” model trained on synthetic data with a method it calls reinforcement learning for calibrated decisions, and keeps the weights and the architecture closed ([TypeSafe AI, 2026b, 2026c, 2026a](#); [TechCrunch, 2026a](#)). Open-weight relatives followed within weeks. Kev attaches a rank-16 LoRA adapter and a small pointer head to Qwen3.5 and Qwen3.8 bases at 0.8B to 27B parameters, ships under Apache-2.0 with a server that speaks the same protocol, and reports accuracy close to Jev’s on held-out classification sets while noting that Jev’s training data is unknown ([Palmer, 2026](#)). Cloudflare’s Clef and Clef-flash freeze Qwen3.8-27B and Qwen3.5-9B and train a routing head with low-rank adapters, are served on Workers AI with vendor-reported median latencies of about 209 and 39 milliseconds against about 524 for Jev, and are likewise Jev-compatible ([Cloudflare, 2026](#)); Perplexity’s pplx-decider fine-tunes Qwen3.8-27B and reports a small lead over Jev on an eleven-task panel ([Perplexity AI, 2026](#)); Amazon released a 2B decider of its own ([TechCrunch, 2026b](#); [The New Stack, 2026](#)). Two smaller systems take different routes. Laya, from Convai Innovations, puts a decision head on encoder models rather than on a generative base, ModernBERT-large at 421M parameters and mmBERT-base at 322M, under Apache-2.0, and reports a few tens of milliseconds per question on a Tesla T4 ([Convai Innovations, 2026](#)). SemIf, formerly OpenJev, trains nothing at all: it reads the logits of the option labels from a frozen open checkpoint such as Qwen3.5-4B in one forward pass, which is precisely the letter-format likelihood scoring we study ([Lee, 2026](#)). In the terms of this paper these systems are the probe and LoRA conversions productized: an open base, frozen or lightly adapted, plus a head that reads probabilities over the allowed answers, exposed through an interface that is exactly our likelihood scoring. What they have not been evaluated on is decision quality against a known optimal policy; their published numbers are label accuracies on classification panels, mostly measured by the vendors themselves. DEAD EYE treats them as one more backend and places them on the same tasks, seeds and normalized scale as the conversions it studies (Section 4.6).

3 The DEAD EYE benchmark

A DEAD EYE task is a generator of small decision problems, each fixed completely by a seed and each paired with an oracle. This section defines the tasks and the score, then the six ways we turn a language model into a policy, and finally what the harness records.

3.1 Decision tasks

A task $\mathcal{T}$ maps a seed $\sigma \in \mathbb{N}$ to an episodic decision problem. Its generator $G_{\mathcal{T}}$ seeds a pseudo-random number generator from σ and draws two things: hidden instance parameters ξ_{σ} , such as the arm means of a bandit, and all the exogenous randomness ω_{σ} the episode can ever consume, such as a table holding the reward of every arm in every round. Given these, an episode is a deterministic finite-horizon problem

$$M_{\sigma} = (\mathcal{S}, \mathcal{A}, \mathcal{A}(\cdot), s_0, f, r, H), \quad s_{t+1} = f(s_t, a_t; \xi_{\sigma}, \omega_{\sigma}), \quad r_t = r(s_t, a_t; \xi_{\sigma}, \omega_{\sigma}),$$

where $\mathcal{A}$ is the set of action labels, $\mathcal{A}(s) \subseteq \mathcal{A}$ the actions that are legal in state s , and H a cap on the number of decisions. The episode ends in a terminal state or after H decisions, and its return is $R(\pi, \sigma) = \sum_t r_t$. Before each decision the policy receives an observation $o_t = (x_t, \mathcal{A}(s_t), \phi(s_t))$: a text rendering x_t of the state, the legal actions, and a numeric feature vector $\phi(s_t)$ that only the language-model-free control and the classical baselines use. Language-model policies also receive a fixed description of the rules. Every task has an oracle $\pi^*(s; \xi_{\sigma})$ that returns an optimal action, chosen by a fixed rule when several actions are optimal.

Because ω_{σ} is drawn before the first decision, two policies run on the same seed face identical potential outcomes: if both pull arm 3 in round 7 of seed 12, both receive the same reward. This is the method of common random numbers ([Law, 2015](#)), and it makes every comparison in this paper a paired comparison over seeds.

The oracles differ in what they know. On gridworld, tic-tac-toe and blackjack the oracle is a function of what the text shows, so a policy that reads the text perfectly can match it. On loan approval the oracle uses a default model that is the same in every instance; the text shows every input of that model but not its coefficients, which a policy has to bring as

prior knowledge or learn from training seeds. On the two bandits the oracle knows the hidden arm means or weights. It is clairvoyant, no policy that learns from feedback can match it in expectation, and a normalized score of 1 is an upper bound rather than a target there; UCB1 provides a realizable reference on the bandit (Section 3.5).

3.2 Normalized score

Raw returns are not comparable across tasks, so we place them on a common scale anchored by two reference policies run on the same seeds:

$$s(\pi, \sigma) = \frac{R(\pi, \sigma) - \bar{R}_{\text{rand}}}{\bar{R}_{\text{orc}} - \bar{R}_{\text{rand}}}, \quad \bar{R}_{\text{rand}} = \frac{1}{|\Sigma|} \sum_{\sigma \in \Sigma} R(\pi_{\text{rand}}, \sigma), \quad \bar{R}_{\text{orc}} = \frac{1}{|\Sigma|} \sum_{\sigma \in \Sigma} R(\pi^*, \sigma),$$

where Σ is the set of evaluation seeds and π_{rand} chooses a legal action uniformly at random. The anchors are means over Σ , so the random and the oracle policy average exactly 0 and 1, while a single episode can score below 0 or above 1. Every task configuration, for example the ten-armed bandit or the sign-flipped loan task, gets its own anchors from its own run. For each cell we report the mean and the interquartile mean of s over episodes with a bootstrap confidence interval, and we summarize a model and method by the unweighted mean over the six tasks. Because s is an affine function of return, it preserves every comparison that return supports; regret, success and win rates are reported alongside. The scaling has a price: when the two anchors lie close together, as in blackjack, the per-episode spread of s is large, and statistical power suffers (Section 4.5).

3.3 Tasks

Table 1 summarizes the seven tasks; Appendix B lists every parameter and distribution, and Appendix A shows a prompt for each.

The seven tasks with the parameters of the main sweep. “Decisions” is the number of decisions per episode, which varies with the policy on gridworld, tic-tac-toe and blackjack.

Task	Decision structure	Oracle	Headline metric	Decisions
bandit	exploration and exploitation without state; 5 arms	best true arm (clairvoyant)	pseudo-regret	50
contextual_bandit	learning an abstract rule in context; 3 options	best option for the context (clairvoyant)	regret	30
loan	one-shot tabular decisions with semantic features; shift variants	approve iff expected value is positive	profit, regret	20
gridworld	deterministic multi-step planning on a 6×6 grid	first step of a shortest path	success, excess steps	≤ 24
tictactoe	adversarial, perfect information; random opponent	expectimax against the random opponent	win rate	≤ 5
blackjack	risk under uncertainty; hit or stand; 20 hands per episode	hit/stand basic strategy	mean return per hand	≥ 20
support	applying a written six-rule policy to a customer’s message; 6 actions	the policy’s rules	correct-decision rate, wasted payout	20

3.3.0.1 Multi-armed bandit (bandit). The agent plays K Bernoulli arms for T rounds ($K = 5, T = 50$ in the main sweep; $K = 10, T = 100$ in a harder variant). Following Krishnamurthy *et al.* (2024), one arm a^* , chosen uniformly, has mean $\mu_{a^*} = 0.5 + \Delta/2$ and every other arm has $\mu_a = 0.5 - \Delta/2$, with gap $\Delta = 0.2$. The generator pre-draws $Y_{t,a} \sim \text{Bernoulli}(\mu_a)$ for every round and arm, and pulling arm a_t in round t pays Y_{t,a_t} . The observation gives the round number and, by default, a summary with the number of pulls and the average reward of each arm; the raw format lists the full history of (arm, reward) pairs instead. The oracle pulls a^* every round. The headline

metric is pseudo-regret,

$$\text{Reg}_T = \sum_{t=1}^T (\mu_{a^*} - \mu_{a_t}),$$

and we also log the fraction of suboptimal pulls and the share of the last ten pulls that went to a^* . The feature vector holds each arm’s share of past pulls and its empirical mean, set to 0.5 for arms not yet pulled. **3.3.0.2 Contextual bandit (contextual_bandit).** Each of $T = 30$ rounds shows a context $x_t \in \mathbb{R}^d$ with $d = 4$ and offers $A = 3$ options. Hidden weights $w_a \sim \mathcal{N}(0, \sigma_w^2 I_d)$ with $\sigma_w = 2$ are drawn for each episode, contexts are drawn from $\mathcal{N}(0, I_d)$ and rounded to two decimals, and option a succeeds with probability

$$p_{t,a} = \text{sigmoid}(w_a^\top x_t / \sqrt{d}),$$

with outcomes again pre-drawn for every round and option. The observation shows the current context and the last ten rounds as (context, choice, reward) triples. The features have abstract names (x1 to x4) and the weights change every episode, so prior knowledge cannot help: the only route above chance is to infer the rule from the history in the prompt. The oracle chooses $\arg \max_a p_{t,a}$, regret is $\sum_t (\max_a p_{t,a} - p_{t,a_t})$, and the feature vector is the current context alone. **3.3.0.3 Loan approval (loan).** An episode presents $N = 20$ independent applicants, each with an age, an annual income I , a debt-to-income ratio D , a credit score C , years in current employment E , a requested amount L and a loan purpose. Incomes and amounts are log-normal, the debt ratio follows a Beta(2, 5) distribution and credit scores are normal around 680. The probability that an applicant defaults is

$$p = \text{sigmoid}\left(-1.2 + 2c \frac{C-680}{70} + 2.5 (D - 0.28) + 1.2 \ln(1 + L/I) - 0.08 \min(E, 15) + \rho_{\text{purpose}}\right),$$

with $c = -1$, so that a higher credit score lowers the risk, and a purpose offset ρ between -0.3 (home improvement) and $+0.6$ (small business). Age is displayed but does not enter the model. Approving earns rL with $r = 0.12$ if the loan is repaid and loses λL with $\lambda = 0.6$ on default; denying earns nothing; defaults are pre-drawn. The expected value of an approval is

$$\mathbb{E}[\text{approve}] = (1 - p) rL - p \lambda L,$$

so the oracle approves exactly when $p < r/(r + \lambda) = 1/6$. Each prompt states the gain and the loss in dollars, which reduces the decision to estimating p . The headline metrics are profit and the regret

$\sum_i (\max(\mathbb{E}_i[\text{approve}], 0) - \mathbb{E}_i[\text{chosen}])$, where the chosen action’s expected value is zero for a denial. Two variants test distribution shift with the same generator. Under covariate shift the median income falls from 55k to 35k dollars and the mean credit score from 680 to 600, with the default model unchanged. Under the sign flip $c = +1$: a higher credit score now raises the risk, and a model’s prior knowledge points the wrong way. The feature vector holds the six numeric fields, standardized, and a one-hot purpose. **3.3.0.4 Gridworld navigation (gridworld).** The agent walks on an $n \times n$ grid toward a goal ($n = 6$ in the main sweep; $n = 9$ with 16 walls in a larger variant). Start, goal and n wall cells are distinct cells taken from a random permutation, and the instance is redrawn until the goal is reachable at a shortest-path distance $d^* \geq n$. The legal actions are the moves among up, down, left and right that stay on the grid and avoid walls. Each move costs 0.02 and reaching the goal pays 1, so a successful episode of t moves returns $1 - 0.02 t$ and the oracle returns $1 - 0.02 d^*$; the episode stops at the goal or after $H = 4n$ moves. The observation is an ASCII map with the coordinates of the agent and the goal. The oracle takes the first step of a breadth-first-search shortest path, breaking ties in the fixed order up, down, left, right. We report success and excess steps, $t - d^*$ on success and $H - d^*$ otherwise. The feature vector holds the normalized coordinates of agent and goal and the wall mask. **3.3.0.5 Tic-tac-toe (tictactoe).** The agent plays a full game against a fixed opponent that moves uniformly at random in the main sweep and by minimax in a variant. It plays X, and moves first, on even seeds, and O on odd seeds. The board is shown with empty cells numbered 1 to 9, and the legal actions are the empty cells. A win pays $+1$, a draw 0 and a loss -1 . Against the random opponent the oracle is expectimax: it maximizes the expected

outcome against uniformly random replies and, among moves of equal expected value, takes an immediate win when one is available and otherwise the lowest-numbered cell. Against the minimax opponent the oracle is minimax with depth-aware values, a terminal win being worth $1 + e$ and a loss $-(1 + e)$ with e the number of empty cells left, so that it prefers fast wins and slow losses. That opponent breaks ties between equally good moves at random, with a generator seeded by the episode seed, so that a deterministic policy meets many different games, and no policy can beat it. The headline metric is the win rate, with draws and losses logged; the feature vector encodes the board from the agent’s point of view. **3.3.0.6 Blackjack (blackjack).** An episode is a sequence of H hands against a dealer ($H = 1$ by default and $H = 20$ in the sweep, where the observation also shows the hand number and the running total); each hand follows the Gymnasium convention (Towers *et al.*, 2025): an infinite deck in which each card is drawn uniformly from $\{1, \dots, 9, 10, 10, 10, 10\}$ (an ace, two to nine, and four ten-valued ranks), a dealer who draws to 17 and stands on soft 17, and a payout of 1.5 for a winning two-card 21. One ace counts as 11 whenever that does not take the hand over 21. The only actions are hit and stand; there is no doubling, splitting, surrender or insurance. Each seed pre-draws two card streams, one for the player and one for the dealer, so a policy’s hits never change the dealer’s cards and every policy faces the same dealer hand. The oracle is the hit/stand basic-strategy table (Baldwin *et al.*, 1956): with a hard total it stands on 17 or more, on 13 to 16 against a dealer card from 2 to 6, and on 12 against 4 to 6; with a soft total it stands on 19 or more and on soft 18 against 2 to 8. For these rules the table coincides with the optimal hit/stand policy, which we checked by dynamic programming over the player’s total, the soft-hand flag and the dealer’s card (Sutton and Barto, 2018). It is optimal only within the hit/stand game, since the task offers no doubling or splitting. The episode return is the sum of the hand outcomes, so bundling H hands divides the standard deviation of the normalized score by about $\sqrt{H}$ without changing the decision problem. The headline metric is the mean return per hand, with win, loss and bust rates logged; the features are the hand total, a soft-hand flag and the dealer’s up card. **3.3.0.7 Customer-support triage (support).** The agent resolves n support tickets in a row ($n = 20$ in the sweep). Each ticket is a customer’s message, such as “I received the package damaged. I need a refund today.”, together with the order facts an agent would see: the item and its value, when it was delivered or how late it is, whether photo evidence is attached, how many claims the customer has made before, the customer’s tier and whether a replacement is in stock. The task description states the company’s six-rule policy, which covers repeat claimers, billing errors, changed minds, late deliveries, missing deliveries and damaged or wrong items, and the six actions are refund, replace, partial refund, request evidence, escalate and deny. The oracle applies the rules literally, so this task measures whether a model can follow a written policy on a concrete case rather than whether it has a good prior. A correct decision earns $+1$ and a wrong one -1 ; an unwarranted payout costs its value in hundreds of dollars on top, and refusing a claim the policy would have paid costs a further 1 for the lost customer. The headline metric is the share of decisions that match the policy, with the wasted payout logged; the features are the one-hot issue and request plus the scaled order fields.

3.4 Conversion methods

A conversion method turns a language model, or in one case no model at all, into a policy (Table 2). All six share one prompt skeleton, so that two conditions differ only in the factor under study. The first of them, the free reply, is not a conversion at all: it is the assistant as shipped, answering the situation in its own words, and it is the reference against which the cost and the reliability of every real conversion are measured.

The six conversion methods. Only the two generation methods can produce an illegal action.

Method	What it uses	Training data	Backends
prompt_free	the unconverted assistant’s free reply, with an action extracted when the text contains one	none	hf, openai
prompt_generate	generated text, parsed into an action (zero-shot, few-shot or chain-of-thought)	none	hf, openai
prompt_score	log-likelihood of each legal label as a continuation of the prompt	none	hf (exact), openai (first token), decision (a decision model’s probabilities)

Method	What it uses	Training data	Backends
probe	last-token hidden state at one layer, with a logistic-regression head	oracle-labeled states from training seeds	hf
lora_sft	LoRA adapters trained by behavior cloning, then exact scoring	the same states	hf
feature_probe	no language model: the same head on the task’s numeric features	the same states	any

3.4.0.1 Prompt skeleton. The system message is the task description. The user message holds the observation text, a line `Legal actions:` that lists the legal labels, and an answer instruction. In the *name* format the labels are the action names themselves (`arm_3`, `left`, `7`). In the *letter* format the legal actions receive the letters A, B, C and so on, in order, and one line per letter states its action. The instruction asks for a single line `Action: <label>` in generation mode, for at most three short sentences of reasoning followed by that line in chain-of-thought mode, and for the label alone in scoring mode. Few-shot demonstrations are complete user and assistant turns placed before the final user message; each shows a state from a training seed, reached after zero to four oracle steps, together with the oracle’s action. Instruction-tuned checkpoints receive the messages through their chat template. Base checkpoints receive a plain rendering in which user and assistant turns are introduced by lines reading `Input:` and `Output:`, even if their tokenizer ships a chat template. The model sees its own past decisions only where the observation itself shows them, as in the bandit summaries and the contextual-bandit history. Appendix A reproduces every variant verbatim.

3.4.0.2 Free reply (prompt_free). The user message ends with “What do you do? Answer in your own words.” instead of a format instruction, and the model decodes up to 128 tokens. The parser then looks for an action in the reply exactly as it does for generation; a reply that names no legal action, or several without committing to one, counts as illegal. This is the “before” condition: it records how long an unconverted assistant takes, how many tokens it spends, and how often its answer contains a usable decision at all.

3.4.0.3 Free-form generation (prompt_generate). The model decodes greedily up to 16 new tokens, 256 with chain-of-thought and 2048 for long thinking budgets; one ablation samples at temperature 0.7 instead. Greedy decoding overrides any repetition penalty that a checkpoint ships with, so base and instruct checkpoints decode the same way. A parser then maps the text to a label. It removes `<think>` blocks, including an unterminated block left when thinking exhausted the budget, and reads the last line of the form `Action: ...`, JSON-style lines included, whose content matches a legal label once quotes, brackets and punctuation are stripped. Single-character labels (letters and tic-tac-toe cells) must match exactly, apart from a leading word such as *option* or *cell*; longer labels may also match as a prefix, as a whole word, or by close spelling (a similarity ratio of at least 0.8). Without such a line the parser takes the last line that matches a label under the same rules and, for multi-character labels only, finally the last label mentioned anywhere in the text. Anything else is a format failure.

3.4.0.4 Likelihood scoring (prompt_score). Scoring never asks the model to write anything. For each legal label c , tokenized on its own as $(c_1, \dots, c_m)$, the score is its log-probability as a continuation of the rendered prompt,

$$\ell(c) = \sum_{j=1}^m \log p_{\theta}(c_j \mid \text{prompt}, c_{<j}), \quad \ell_{\text{norm}}(c) = \ell(c)/m,$$

and the policy takes $\arg \max_c \ell(c)$, or $\arg \max_c \ell_{\text{norm}}(c)$ in the length-normalization ablation, with ties going to the first label. All labels are scored in one batched forward pass; this is the cloze-style scoring used for multiple-choice evaluation (Holtzman et al., 2021; Robinson and Wingate, 2023). When the tokens of one label are a strict prefix of another’s, as for `arm_1` and `arm_10` under a tokenizer that splits digits, every label is scored with an end-of-sequence token appended; otherwise the longer label could never win. The policy cannot produce an illegal action. Models behind an OpenAI-compatible API expose only the top log-probabilities of the first generated token, so for them each label is scored by its first token. That is exact only when the labels begin with distinct tokens, which is why we run such models with letter labels and report them separately.

3.4.0.5 Hidden-state probe (probe). The probe reads a decision from the model’s representation of the prompt without generating anything. Let $h_{\ell}(x)$ be the hidden state of the last prompt token after block ℓ of a model with B blocks, counting the embedding output as $\ell = 0$. We use the middle layer, $\ell = \lfloor B/2 \rfloor$, and in an ablation the last layer, $\ell = B$. A multinomial logistic regression with an L_2 penalty (inverse strength $C = 1$), fitted with scikit-learn (Pedregosa et al., 2011), is trained on the standardized h_{ℓ} to predict the

oracle’s action. The training states come from the first N_{train} training seeds, 100 in the main sweep and 10, 40 or 100 in the data-budget ablation. A behavior policy that takes the oracle’s action with probability 0.7 and a uniformly random legal action otherwise visits them, so that states off the oracle’s path are covered too, and every visited state is labeled with the oracle’s action. Collection would stop at 50000 states, a cap that none of our configurations reaches; the number of training episodes actually used is logged. For the two shifted loan variants the training episodes come from the unshifted task, so adapted policies are evaluated out of distribution there; few-shot demonstrations follow the same rule. At decision time the probe takes the legal action with the highest predicted probability. It embeds the generation-mode prompt. **3.4.0.6 LoRA behavior cloning (lora_sft).** LoRA adapters (Hu et al., 2022) of rank 16 with scale $\alpha = 32$ and dropout 0.05 are attached to every linear layer and trained for two epochs on the probe’s training states, with AdamW, a learning rate of 2×10^{-4} , batch size 8, no weight decay and gradient norms clipped at 1. Each example is the scoring-mode prompt followed by the oracle’s label and an end-of-sequence token, and the cross-entropy loss covers only those target tokens. After training the adapters are merged into the weights, except for 4-bit models, which are trained as QLoRA (Dettmers et al., 2023) and kept unmerged. The policy then acts by exact scoring with the same prompt. Every LoRA cell starts from a fresh copy of the pre-trained weights. **3.4.0.7 Feature probe (feature_probe).** The control uses no language model. It is the same logistic regression, trained on the same states, with the task’s feature vector $\phi(s)$ in place of h_ℓ . What it controls for differs by task, and Section 6.3 discusses how to read it. **3.4.0.8 Illegal actions.** Only the two generation methods, the free reply and free-form generation, can fail to yield a legal action. When the output cannot be mapped to a legal label, the runner substitutes a legal action drawn uniformly at random, from a generator seeded by the episode seed, and marks the decision as illegal. The substituted action counts toward the return but never toward oracle agreement. The illegal-action rate, the share of decisions marked illegal, is a secondary outcome. The harness can instead take the first legal action or end the episode; we use random substitution throughout, so that a format failure is scored like an uninformed decision rather than a forfeit.

3.5 Baselines

Three policies need no language model. The random policy draws a legal action uniformly, from one generator per cell. The oracle is π^* . On the bandit we also run UCB1 (Auer, Cesa-Bianchi and Fischer, 2002), which pulls each arm once and then always the arm that maximizes

$$\hat{\mu}_a + \sqrt{2 \ln(t+1)/n_a},$$

where t is the number of completed rounds, n_a the number of pulls of arm a and $\hat{\mu}_a$ its mean reward so far. UCB1 is the textbook reference, not the strongest algorithm for a 50-round horizon. Its normalized score, [result pending: [s of UCB1 on bandit with CI]], marks what a non-clairvoyant learner with a regret guarantee achieves on this task.

3.6 Logging and provenance

A run crosses the task configurations, models and methods of a configuration file into cells, and evaluates each cell on every evaluation seed. Each cell writes three files to `results/<run>/<task>/<model>/<method>/`. The first, `summary.json`, holds the aggregates (mean and standard deviation of the return and of every task metric, illegal-action rate, oracle agreement, latency and prompt and completion tokens per decision) together with the full model and method specification, the model’s precision and parameter count, the seeds, the task parameters used for training, statistics from the training phase such as probe accuracy, LoRA loss before and after training and the number of training episodes used, timings and the harness version. The second, `episodes.jsonl`, has one record per episode: seed, return, task metrics, illegal-action count, oracle agreement, latency and token counts. The third, `steps.jsonl`, has one record per decision: legal actions, the oracle’s action, the parsed and the executed action, whether the decision was legal, the reward, latency, token counts, the raw model output truncated to 2000 characters, and method-specific extras such as the score of every label or the probe’s class probabilities. Each run directory also holds `run_manifest.json`, with the configuration, the list of cells, start and finish times, the Python, PyTorch and Transformers versions, whether CUDA was available and the GPU name. A cell whose summary already exists is skipped, so runs can be resumed and split across machines, and `deadeye` report rebuilds every result table and figure in this paper from these files.

4 Experimental setup

4.1 Model ladders

A scale effect means something only within a family trained with one recipe, so we evaluate ladders rather than a collection of popular models (Table 3). Qwen2.5 (Yang *et al.*, 2024) is the primary ladder: seven dense sizes from 0.5B to 72B, released together, with base and instruct checkpoints at every size. Qwen3 (Yang *et al.*, 2025) adds six dense sizes whose post-trained checkpoints switch thinking on and off inside the same weights; its base checkpoints stop at 14B. SmolLM2 (Ben Allal *et al.*, 2025) and Gemma 3 (Gemma Team, 2025) extend the low end to 135M and 270M parameters. Two fully open ladders serve as controls in a separate configuration: Pythia (Biderman *et al.*, 2023), eight base-only sizes trained on the same data in the same order, and OLMo 2 (Walsh *et al.*, 2025), with open data and base and instruct checkpoints.

We leave out ladders that would put different kinds of model on one axis. The Llama 3 herd (Grattafiori, Dubey, *et al.*, 2024) covers 8B, 70B and 405B, and the smaller Llama sizes come from later releases; the DeepSeek-R1 distillations sit on base models from two families (Guo *et al.*, 2025); and mixture-of-experts models would force a choice between active and total parameters. Newer dense ladders, such as Olmo 3 (Team Olmo, Ettinger, *et al.*, 2025), can be added through the model catalog once their repository identifiers are verified.

Model ladders. “Released” lists the checkpoints that exist and “Run” those that our configurations evaluate. Weights are bfloat16 unless marked int4 (4-bit NF4). Appendix F gives the verified repository identifiers.

Family	Sizes	Released	Run	License
Qwen2.5	0.5B, 1.5B, 3B, 7B, 14B, 32B, 72B	base and instruct at every size	base and instruct up to 14B; instruct at 32B and 72B (int4)	Apache-2.0; 3B under the Qwen Research License, 72B under the Qwen License
Qwen3	0.6B, 1.7B, 4B, 8B, 14B, 32B	post-trained at every size; base up to 14B	post-trained with thinking off (32B in int4); thinking on at 4B and 8B	Apache-2.0
SmolLM2	135M, 360M, 1.7B	base and instruct	instruct	Apache-2.0
Gemma 3	270M, 1B, 4B, 12B, 27B	pre-trained and instruct	instruct (27B in int4)	Gemma Terms of Use
Pythia	70M, 160M, 410M, 1B, 1.4B, 2.8B, 6.9B, 12B	base only	deduplicated base	Apache-2.0
OLMo 2	1B, 7B, 13B, 32B	base and instruct	base and instruct up to 13B; instruct at 32B (int4)	Apache-2.0

Models up to 14B run in bfloat16. The largest rungs (Qwen2.5 32B and 72B, Qwen3 32B, Gemma 3 27B and OLMo 2 32B) are loaded with 4-bit NF4 weights through bitsandbytes (Dettmers *et al.*, 2023), and the quantization ablation adds 8-bit weights with LLM.int8() (Dettmers *et al.*, 2022), so the top of each ladder mixes scale with quantization; Section 5.5 measures the cost of 4-bit weights at sizes where both precisions fit, and every table marks the precision of each row. The parameter axis is the total parameter count from the model catalog, which quantized rows report as well. Embeddings take a large share of the smallest models, about a quarter of Qwen2.5-0.5B’s 0.49B parameters according to its model card, and more in models with large vocabularies, so we repeat the slope fits with non-embedding counts where they are published.

4.2 Factors and levels

Table 4 lists the factors we vary. The main sweep crosses the ladders with seven method configurations (the free reply, generation, generation with chain-of-thought, scoring, probe, LoRA and the feature probe) on the seven tasks. The control configuration does the same for Pythia and OLMo 2 without chain-of-thought. Five ablation blocks vary one factor each: quantization on Qwen2.5-Instruct at 1.5B, 7B and 14B in three precisions; prompting and decoding on the same three models; thinking on Qwen3 at 1.7B, 4B, 8B and 14B with the mode on and off and three token budgets; the

adaptation data budget and probe layer on Qwen2.5-Instruct at 0.5B, 1.5B and 7B; and task structure over thirteen task configurations, including the shifted loan variants.

Factors, their levels in the configuration files, and the hypotheses they test. “Sweep”, “controls” and “ablations” are `configs/sweep_gpu.yaml`, `configs/sweep_controls.yaml` and the five files under `configs/ablations/`.

Factor	Levels	Configuration	Tests
Parameter count	Qwen2.5 0.5B to 72B; Qwen3 0.6B to 32B; SmolLM2 135M to 1.7B; Gemma 3 270M to 27B; Pythia 70M to 12B; OLMo 2 1B to 32B	sweep, controls	H1
Instruction tuning	base vs. instruct at equal size: Qwen2.5 0.5B to 14B; OLMo 2 1B to 13B	sweep, controls	H2
Conversion method	<code>prompt_generate</code> , <code>prompt_score</code> , <code>probe</code> , <code>lora_sft</code> , <code>feature_probe</code>	all	H3, H4
Quantization	bf16, int8 and int4 at 1.5B and 7B; bf16 and int4 at 14B	ablations	H5
History format	bandit history as a per-arm summary vs. the raw list	ablations	H6
Reasoning budget	direct answer (16 tokens) vs. chain-of-thought (256); Qwen3 thinking off vs. on (up to 2048 tokens)	sweep, ablations	H7
Prompt details	names vs. letters; 0 vs. 3 demonstrations; temperature 0 vs. 0.7; raw vs. length-normalized scores	ablations	secondary scores
Adaptation data	probe and LoRA trained on 10, 40 or 100 episodes	ablations	H3
Task difficulty	5 vs. 10 arms; grid 6 vs. 9; random vs. minimax opponent; loan shift none, covariate or sign flip	ablations	H1, H3

4.3 Seeds and data splits

Evaluation uses seeds 0 to 99 for every task configuration except tic-tac-toe, which uses seeds 0 to 299 because a game’s outcome is -1 , 0 or $+1$; blackjack keeps seeds 0 to 99 but plays twenty hands per episode for the same reason. Probe and LoRA training states and few-shot demonstrations come only from seeds 100000 to 100099, and the harness rejects any configuration in which training and evaluation seeds overlap. Every model and method sees the same evaluation seeds, so all comparisons within a task configuration are paired. The CPU pilot, in which prompts are debugged, evaluates seeds 5000 to 5019 with training seeds 100000 to 100039, so no confirmatory instance is seen while prompts are being edited.

4.4 Hardware and software

Models run through the Transformers library ([Wolf et al., 2020](#)) with LoRA through PEFT, probes through scikit-learn ([Pedregosa et al., 2011](#)) and quantized weights through bitsandbytes; the run manifest records the Python, PyTorch, Transformers, PEFT, scikit-learn, NumPy and bitsandbytes versions, CUDA availability and the GPU name, so every result directory carries its own software record. [\[result pending: \[hardware used for the sweep\]\]](#) With greedy decoding and seeded instances, `prompt_score`, `probe` and greedy `prompt_generate` reproduce bit for bit on the same hardware and library versions; across hardware we expect agreement within confidence intervals, not identical values. Latencies are wall-clock times of the model’s forward passes per decision on this hardware.

4.5 Statistical protocol

The unit of analysis is the episode. The primary outcome is the normalized score s of Equation (2); the secondary outcomes are the illegal-action rate, oracle agreement, the task metrics, and latency and tokens per decision. For each cell we report the mean and the interquartile mean of s over episodes and a 95% percentile bootstrap interval from 2000 resamples of episodes ([Efron and Tibshirani, 1993](#); [Agarwal et al., 2021](#)).

Two cells of the same task configuration are compared through the paired differences $\delta_\sigma = s(\pi_A, \sigma) - s(\pi_B, \sigma)$ over their shared seeds. The test statistic is $|\bar{\delta}|$, its null distribution comes from 5000 random sign flips of the δ_σ , and the two-sided p -value is $(1 + \#\{k : |\bar{\delta}^{(k)}| \geq |\bar{\delta}|\})/5001$. Effect sizes are paired Cohen’s $d = \bar{\delta}/\text{sd}(\delta)$. Within each hypothesis, the tests named in its pre-registered criterion form one family and are corrected with Holm’s step-down procedure (Holm, 1979); `deadeye.compare` takes one family of pairs per call and applies the correction within it.

Scale is summarized by the least-squares fit

$$\bar{s}_c = a + b \log_{10} N_c$$

over the cells c of one family, method and task, where $\bar{s}_c$ is the cell mean of s and N_c the parameter count; b is the change in normalized score per tenfold increase in parameters. Confidence intervals for b , for differences of slopes (H1b) and for ratios of slopes (H4b) come from refitting after resampling episodes within every cell, 1000 times; `deadeye.report` writes the fits to `slopes.csv`, including a fit on the cross-task mean score per model, and `deadeye.report.slope_ratio` forms the ratios.

The smallest effect of interest is $\Delta = 0.10$ in s . For a paired comparison in which the per-episode differences have standard deviation σ_δ , the number of episodes needed for power $1 - \beta$ at level α is

$$n = \left\lceil \left((z_{1-\alpha/2} + z_{1-\beta}) \sigma_\delta / \Delta \right)^2 \right\rceil,$$

which `deadeye.metrics.episodes_needed` implements. With $\alpha = 0.05$ and power 0.8 this gives 32 episodes at $\sigma_\delta = 0.2$ and 126 at $\sigma_\delta = 0.4$, so 100 episodes suffice when σ_δ is at most about 0.36. The pilot estimates of σ_δ are [result pending: [per-task SD of paired differences from the pilot]]. Tasks whose anchors lie close together have a much larger per-episode spread of s . For blackjack the per-hand standard deviation of the return is close to 1, so the standard deviation of s is roughly the reciprocal of the anchor span [result pending: [span $\bar{R}_{\text{orc}} - \bar{R}_{\text{rand}}$ on blackjack]], which is why the sweep bundles twenty hands into each of its 100 blackjack episodes and runs 300 tic-tac-toe games: the standard deviation of an episode mean falls with the square root of the number of hands. The minimum detectable difference under this design is [result pending: [per task, at power 0.8]], and where it exceeds 0.10 we report it next to the estimate rather than reading a non-significant result as a null.

A cell is excluded only if its model fails to load or its run crashes, never because of its outcome, and Appendix C lists every excluded cell.

4.6 Decision models

A fourth backend, `decision`, speaks the System One protocol of Jev and its open-weight relatives (TypeSafe AI, 2026a; Palmer, 2026; Cloudflare, 2026): the rendered task description and observation go out as the state, the legal actions as the criteria of a single choice question, and the returned probabilities are used exactly as the log-likelihoods of `prompt_score`. Table 5 lists the systems and what is verified about each from its maker’s cards, announcements and repositories; every performance figure quoted there is the maker’s own. The configuration `configs/decision_models.yaml` evaluates them next to the same Qwen3.5 base checkpoints converted by our own scoring, probe and LoRA methods, and next to the small Qwen2.5 and Qwen3 checkpoints of the ladders. Only scoring applies to a decision model; it cannot generate, so the free-reply and generation conditions are skipped for it. Two rows double as validations: SemIf’s readout is our letter-scoring condition, so the two should agree on the same checkpoint up to prompt wording, and Kev on a Qwen3.5 base differs from our LoRA conversion of the same base only in its training data, which isolates what the training target contributes. Every system is reduced to the same numbers: normalized score per task with its mean and worst task, oracle agreement, calibration of the probability assigned to the chosen action against the oracle (expected calibration error and Brier score), latency per decision (median and 95th percentile, with network time reported separately for hosted systems), tokens, parameters and openness.

Decision models compared with the open-weight conversions. Sizes, bases, licenses and training descriptions are from the makers’ cards and announcements; “how run” is our access path. Jev’s base and training data are undisclosed.

System	Maker, license	Base and size	How it decides (as reported)	How run
Jev (TypeSafe AI, 2026b)	TypeSafe AI; closed	undisclosed	“System One” model; reinforcement learning for calibrated decisions on synthetic data	hosted API
Kev (Palmer, 2026)	J. Palmer; Apache-2.0	Qwen3.5 0.8B/4B/9B-Base with rank-16 LoRA and a pointer head; Qwen3.8-27B full fine-tune	trained on a 10k-example decision set plus generated policy examples; System One server	local server
Clef, Clef-flash (Cloudflare, 2026)	Cloudflare; Apache-2.0	Qwen3.8-27B, Qwen3.5-9B, frozen	routing head with low-rank adapters; Jev-compatible	Workers AI
pplx-decider v1.1 (Perplexity AI, 2026)	Perplexity; Apache-2.0 reported, gated	Qwen3.8-27B fine-tune	Decisions API with its own schema	adapter
Laya (Convai Innovations, 2026)	Convai Innovations; Apache-2.0	ModernBERT-large 421M; mmBERT-base 322M	encoder with a decision head; Choice, Score and Noul primitives	local package
SemIf (Lee, 2026)	T. Lee; MIT code	frozen Qwen3 0.6B, MiniCPM5 2B, Qwen3.5-4B	option-logit readout, no training	local; also reproduced by our letter scoring

This comparison is exploratory and was added after the pre-registration: the hosted model’s training data and architecture are unknown, so contamination cannot be checked, its latency includes network time and is reported separately from the local measurements, and we state no hypothesis about it. What it can show is whether a conversion that anyone can reproduce on open weights reaches the decision quality of the hosted product on tasks with a known optimum, and at what cost.

4.7 Pre-registration

The hypotheses, predictions and falsification criteria (Appendix C) are written before the main sweep and frozen, together with the prompts and the analysis code, at the git tag `prereg-v1`, which is created after the CPU pilot. Prompts may change only during the pilot and only against pilot seeds. Every run after the tag is confirmatory, and any departure from the frozen design is recorded with its date and reason in the deviations table of `docs/preregistration.md` and in Appendix C. [TODO: Commit hash and date of the prereg-v1 tag; deviations, if any.]

5 Results

We take the hypotheses in their pre-registered order and test each against its criterion (Appendix C). Unless stated otherwise, a number is a mean of s over 100 episodes, 300 for blackjack, with a 95% bootstrap interval, and every comparison is paired over seeds. Table 6 gives the main sweep by conversion method; Figure 1 shows the same cells against parameter count.

[Table placeholder: generated from the results by deadeye report] [Table placeholder: generated from the results by deadeye report] [Table placeholder: generated from the results by deadeye report] [Table placeholder: generated from the results by deadeye report]

[TODO: deadeye report writes each table as a captioned table float (pandas to_latex with caption and label); a float cannot be input inside adjustbox within this table, so the build will fail once the files exist. Emit bare tabular environments from report.py, or input the files at top level.]

5.1 H1: scale under generation

5.1.0.1 Prediction. Under `prompt_generate`, normalized score rises roughly linearly in the logarithm of parameter count in every family with at least four sizes (H1a), and the slope is steeper on gridworld and tic-tac-toe than on loan approval (H1b).

[Figure placeholder: generated from the results by deadeye report]

Normalized decision quality (random = 0, oracle = 1) versus parameter count, one panel per task, one line per conversion method and model family. Error bars are 95% bootstrap confidence intervals over episodes; dotted line: UCB1 on the bandit; dash-dotted line: the feature probe, which uses no language model.

5.1.0.2 Result. Under generation the fitted slope (Equation (9)) is [result pending: [b with 95% CI]] per tenfold increase in parameters for Qwen2.5, [result pending: [b , CI]] for Qwen3, [result pending: [b , CI]] for Gemma 3, [result pending: [b , CI]] for Pythia and [result pending: [b , CI]] for OLMo 2. SmolLM2 has only three sizes and is plotted but not tested. H1a is therefore [result pending: [supported / falsified, naming any family whose interval includes 0]]. Averaged over gridworld and tic-tac-toe, the slope exceeds the loan slope by [result pending: [$b_{\text{plan}} - b_{\text{loan}}$ with CI]], and H1b is [result pending: [supported / falsified]]. The curves are [result pending: [smooth / show a threshold near N parameters on task]], and the 4-bit points at the top of each ladder lie [result pending: [on / below]] the trend of the bfloat16 points. [TODO: Slopes from report/sweep_gpu/episodes.csv and report/sweep_controls/episodes.csv by refitting Equation (9) on bootstrap resamples; cell means and intervals from report/sweep_gpu/cells.csv.]

5.2 H2: instruction tuning

5.2.0.1 Prediction. At equal size, instruction-tuned checkpoints score higher than base checkpoints under `prompt_generate` at no fewer than 80% of sizes (H2a), while under probe and `lora_sft` the instruct-minus-base gap lies within ± 0.05 (H2b). **5.2.0.2 Result.** Under generation the instruct checkpoint leads at [result pending: [k of 7]] Qwen2.5 sizes, [result pending: [k of 5]] Qwen3 sizes and [result pending: [k of 3]] OLMo 2 sizes, with paired differences between [result pending: [smallest]] and [result pending: [largest difference]]; H2a is [result pending: [supported / falsified]]. [result pending: [Much / little]] of this gap is formatting: base checkpoints produce no parseable action in [result pending: [illegal-action rate, base]] of decisions, against [result pending: [illegal-action rate, instruct]] for instruct checkpoints. Under the probe the mean gap is [result pending: [gap with CI]] and under LoRA [result pending: [gap with CI]], so H2b is [result pending: [supported / falsified]]. Scoring, which also removes formatting, gives a gap of [result pending: [gap with CI]]. [TODO: Paired tests per size and task with deadeye compare, e.g. `deadeye compare results/sweep_gpu --env loan --a Qwen__Qwen2.5-7B-Instruct/prompt_generate -b Qwen__Qwen2.5-7B/prompt_generate`; pairwise_methods.csv holds only method pairs within one model. Illegal-action rates from cells.csv.]

5.3 H3: adaptation versus scale

5.3.0.1 Prediction. In distribution, the best `lora_sft` or probe model with at most 1.7B parameters matches or beats a `prompt_generate` model of the same family about ten times larger on at least five of the seven tasks (H3a); under the covariate-shifted and sign-flipped loan variants, the small adapted model loses more relative to its in-distribution score than the large zero-shot model does (H3b). **5.3.0.2 Result.** The adapted small models match or beat their tenfold larger counterparts on [result pending: [k of 6]] tasks, for example [result pending: [Qwen2.5 1.5B with LoRA against Qwen2.5 14B with generation: paired difference and CI per task]]; H3a is [result pending: [supported / falsified]]. Adaptation helps most on [result pending: [tasks]] and least on [result pending: [tasks]], and in the data-budget ablation LoRA [result pending: [reaches the larger model's score after n training episodes / does not reach it with 100]], the probe [result pending: [likewise]]. The feature probe, which uses no language model, reaches [result pending: [s by task]] on the same training states (Section 6.3). Under shift, the drop from the in-distribution score is [result pending: [drop with CI]] for the small adapted model and [result pending: [drop with CI]] for the large zero-shot model, an interaction of [result pending: [difference with CI]]; H3b is [result pending: [supported / falsified]]. [TODO: Pairing rule: in each family, the rung nearest to ten times the adapted model (Qwen2.5 1.5B with 14B, 0.5B with 7B); confirm against the frozen pre-registration. Cross-model and cross-configuration pairs via deadeye compare (pairs may name

different task configurations, matched by seed); ablation curves from report/ablations/cells.csv. The shifted loan cells train adapted methods and demonstrations on the unshifted task (train_params in configs/ablations/tasks.yaml); check summary.json (train_env_params) for every H3b cell.]

5.4 H4: scoring versus generation

5.4.0.1 Prediction. Under prompt_score the illegal-action rate is zero by construction (H4a), and its scale slope is less than half of the slope under prompt_generate (H4b).

[Figure placeholder: generated from the results by deadeye report]

Illegal or unparseable action rate under free-form generation versus parameter count.

5.4.0.2 Result. H4a holds by construction; across [result pending: [number]] scored decisions none was illegal. Under generation the illegal-action rate is [result pending: [rate]] for the smallest model and [result pending: [rate]] for the largest (Figure 2), and is highest on [result pending: [task]]. The ratio of slopes $b_{\text{score}}/b_{\text{gen}}$ is [result pending: [ratio with CI]], so H4b is [result pending: [supported / falsified]]; in Qwen2.5, scoring closes [result pending: [Z]]% of the gap between the 0.5B and the 72B model under generation. Letter labels change scores by [result pending: [paired difference with CI]] and length normalization by [result pending: [paired difference with CI]] (ablations). [TODO: Illegal-action rates from report/sweep_gpu/cells.csv (illegal_rate) and tables/illegal_rate.tex; slope ratio from episodes.csv; generation against scoring per model from pairwise_methods.csv; format ablations from report/ablations/pairwise_methods.csv.]

5.5 H5: quantization

5.5.0.1 Prediction. Replacing bfloat16 weights with 4-bit weights costs less than 0.05 in s at 7B and 14B and more than 0.10 at 1.5B.

5.5.0.2 Result. The paired bfloat16-minus-int4 difference under generation is [result pending: [difference with CI]] at 1.5B, [result pending: [difference with CI]] at 7B and [result pending: [difference with CI]] at 14B; under scoring it is [result pending: [differences with CI]]. Eight-bit weights cost [result pending: [differences with CI]]. H5 is [result pending: [supported / falsified]]. [TODO: deadeye compare results/ablations --env <task> -a q7b-bf16/prompt_generate --b q7b-int4/prompt_generate, per size, task and method; Holm over the pre-registered family.]

5.6 H6: exploration

5.6.0.1 Prediction. No model under prompting reaches the normalized score of UCB1 on the bandit (H6a), and for models below 2B a per-arm summary of the history beats the raw history (H6b).

5.6.0.2 Result. UCB1 reaches [result pending: [s with CI]] on the bandit. The best prompted cell, [result pending: [model and method]], reaches [result pending: [s with CI]], and [result pending: [number]] of [result pending: [number]] prompted cells have an upper confidence bound below UCB1's score; H6a is [result pending: [supported / falsified]]. Prompted models leave [result pending: [fraction]] of the arms unpulled on average (Section 6.1). For Qwen2.5-1.5B-Instruct, the only model below 2B in the history ablation, the summary-minus-raw difference is [result pending: [difference with CI]], so H6b is [result pending: [supported / falsified]] for the one model it can be tested on. [TODO: Bandit cells from report/sweep_gpu/cells.csv: norm_ci_hi of the prompted cells against norm_mean of _baseline/ucb1. The summary-versus-raw comparison pairs seeds across the bandit and bandit_rawhist configurations and must be computed from report/ablations/episodes.csv.]

5.7 H7: reasoning budget

5.7.0.1 Prediction. Chain-of-thought or thinking mode improves gridworld and tic-tac-toe for models of 4B and above (H7a), does not improve models below 2B (H7b), and lowers latency-normalized score, s divided by latency per decision, at every size (H7c).

5.7.0.2 Result. With chain-of-thought, scores on gridworld and tic-tac-toe change by

[result pending: [mean paired difference with CI]] for models of 4B and above and by [result pending: [difference with CI]] below 2B. Thinking mode changes them by [result pending: [difference with CI]] for Qwen3-4B and [result pending: [difference with CI]] for Qwen3-8B. H7a is [result pending: [supported / falsified]] and H7b is [result pending: [supported / falsified]]. Reasoning multiplies latency per decision by [result pending: [factor]], and latency-normalized score is lower with reasoning at [result pending: [number]] of [result pending: [number]] sizes; H7c is [result pending: [supported / falsified]]. In [result pending: [fraction]] of thinking-mode decisions the reasoning used up the token budget before any action appeared. [TODO: Sweep: deadeye compare results/sweep_gpu --env gridworld --a <model>/prompt_generate_cot --b <model>/prompt_generate. Ablations: Qwen3-4B-think/gen_cot_long against Qwen3-4B-nothink/prompt_generate. Thinking-mode cells with a 16-token budget are dominated by truncation and should not enter H7. Latency from cells.csv (latency_per_decision_s).]

5.8 Small open-weight conversions against decision models (RQ5, exploratory)

5.8.0.1 Setting. Not pre-registered; Section 4.6 and Table 5. Kev 0.8B/4B/9B, Clef-flash and Clef, Laya, pplx-decider and SemIf, with the hosted Jev as a reference row where access was granted, are scored on the seven tasks and set against the same Qwen3.5 bases and the small Qwen2.5 and Qwen3 checkpoints converted by prompt_score, probe and lora_sft. Table 7 is the complete comparison: one row per system, the same numbers for all.

[Table placeholder: generated from the results by deadeye report]

5.8.0.2 Decision quality. On the policy-following tasks (support, loan) the best decision model reaches [result pending: [s with CI]] against [result pending: [s]] for the best small conversion; on the tasks that need in-context learning or planning (contextual bandit, gridworld) the decision models score [result pending: [s]], [result pending: [above / below]] the conversions of the same bases. The per-task tables are in Appendix C. [TODO: Fill from

report/decision_models/system_comparison.csv and the method tables.] **5.8.0.3 Calibration.** The expected calibration error of the probability each system assigns to its chosen action is [result pending: [ECE by system, per task]]; the systems that advertise calibrated decisions are [result pending: [better / no better]] calibrated against a known optimum than likelihood scoring of a raw base model, and the probes trained on oracle labels reach [result pending: [ECE]].

[TODO: Tables calibration_ece and calibration_brier from the report; compare within a task only (Appendix B on tie-breaking).]

5.8.0.4 Speed and cost. Median latency per decision is [result pending: [ms by system]] with 95th percentiles of [result pending: [ms]]; the hosted system’s figure includes a network round trip of about [result pending: [ms]], which we measured separately. Parameters and tokens per decision give [result pending: [the relative cost]]; the makers’ published latency figures (Cloudflare, 2026; Convai Innovations, 2026) are [result pending: [consistent / not consistent]] with ours on our hardware.

5.8.0.5 Validations. SemIf on Qwen3.5-4B and our letter scoring of the same checkpoint agree on [result pending: [share of decisions]] and differ in score by [result pending: [difference with CI]], which bounds the effect of prompt wording. Kev-9B and our LoRA conversion of Qwen3.5-9B-Base differ by [result pending: [difference with CI]] per task, which is the contribution of training on classification corpora versus on the task’s oracle labels.

5.9 Small versus large open-weight models: quality, speed and efficiency (RQ4)

5.9.0.1 Result. Within each ladder, latency per decision grows [result pending: [factor]] from the smallest to the largest rung under scoring and [result pending: [factor]] under generation, while normalized score grows [result pending: [difference]]; the smallest model within [result pending: [margin]] of the largest on the cross-task mean is [result pending: [model]] under [result pending: [method]]. Tokens per decision are [result pending: [by method]], and the estimated floating-point operations per decision ($2 \times$ parameters $\times$ tokens) span [result pending: [range]]. Figure 3 shows the frontier. [TODO: From cells.csv: latency_p50_s, latency_p95_s, prompt and completion tokens, params; the manifest’s GPU for memory.]

[Figure placeholder: generated from the results by deadeye report]

Mean normalized score across tasks versus latency per decision.

Figure 3 places every model and method by its mean s across tasks and its latency per decision. The frontier consists of [result pending: [cells on the frontier]]. Below [result pending: [seconds]] per decision the best cell is [result pending: [model and method]]; the best cell at any latency is [result pending: [model and method]], which is [result pending: [factor]] times slower. These latencies exclude the one-off cost of training probes and adapters, which Section 6.2 adds back. [TODO: From report/sweep_gpu/cells.csv (latency_per_decision_s, prepare_time_s).]

6 Analysis

The hypotheses ask whether an effect exists; the per-decision logs let us ask where it comes from. The analyses below are fixed before the confirmatory runs, and every quantity in them is computed from `steps.jsonl` and `episodes.jsonl`.

6.1 Format failures and decision errors

Each decision under `prompt_generate` falls into one of four classes. A *format failure* produced no legal label and was replaced by a random action; these are the decisions logged as illegal. A *lenient parse* yielded a legal label only through one of the parser’s fallback rules, a bare label on a late line or the last label mentioned anywhere, rather than through an `Action:` line; we recover this class by re-running the parser’s stages on the logged raw output. A *decision error* is a legal action that is not optimal, and the remaining decisions are optimal. We judge optimality against the set of optimal actions, not the single action the oracle logged. Gridworld often has several shortest paths and tic-tac-toe several equally good cells, and every episode is deterministic given its seed and the actions taken, so we replay each episode, recover every visited state and compute its full optimal set.

The four rates, by model, method and task, show how much of the gap between small and large models is formatting. [result pending: [Share of the generation-minus-oracle gap attributable to format failures, by size and task]]. The comparison with scoring can be made decision by decision on loan approval, where the applicants do not depend on earlier actions, so every policy sees the same states; on the other tasks only the first decision of each episode is shared. On those shared states, generation and scoring choose the same legal action in [result pending: [agreement rate by size]] of the decisions where generation produced one. If the two agree on most such decisions, the benefit of scoring in H4 is mostly the removal of format failures; if they do not, scoring also changes which legal action the model prefers.

Chain-of-thought and thinking outputs let us look for the knowing-doing gap of Schmied *et al.* (2026). For each such decision we take the reasoning, either the text before the final `Action:` line or the `<think>` block, and check whether it names an optimal action or states the fact that makes an action optimal: an immediate win or a forced block in tic-tac-toe, a total inside a stand region of the basic-strategy table in blackjack, the shorter way around a wall in gridworld. We then check whether the final action differs. Automatic matching proposes candidates and a stratified sample is annotated by hand [result pending: [sample size and agreement between automatic and manual labels]]. Only the explicit `Action:` line counts as the final action here, because the parser’s last-mention rule could otherwise take the action from the reasoning itself. [result pending: [Rate of decisions with a correct rationale and a wrong action, by size and task]].

On the bandit we separate the failure modes named in prior work. Greediness is the share of arms never pulled (Schmied *et al.*, 2026); frequency bias is the probability of repeating the arm chosen most often so far; and a failure to exploit is choosing, late in an episode, an arm other than the one with the highest empirical mean among arms pulled at least k times (Harris and Slivkins, 2026), with k fixed before the confirmatory runs [TODO: choose k and add it to the pre-registration]. The suboptimal-pull fraction and the share of the last ten pulls on the best arm are logged per episode. [result pending: [The three rates by size and method]].

Errors also have a direction. In blackjack, standing where the table hits is conservative and hitting where it stands is aggressive; we tabulate both by player total, soft or hard hand, and dealer card, to see whether models are systematically risk-averse, as Jia *et al.* (2024) found language models to be in multiple-choice-list experiments [result pending: [conservative and aggressive error rates by size]]. In loan approval the analogous quantity is the approval rate relative to the oracle’s [result pending: [approval rates]]. Under the sign flip we regenerate each applicant from its seed and ask

how many errors follow the usual meaning of the credit score, that is, how many decisions match what the unflipped default model would prescribe [\[result pending: \[share of prior-following errors by size and method\]\]](#).

6.2 Quality versus cost

Latency per decision mixes model size, method and hardware, so we report it next to token counts, which do not depend on hardware. Scoring needs one forward pass over the prompt and the candidate labels, generation adds one decoding step per generated token, the probe needs one forward pass and a logistic regression, and LoRA adds a one-off training cost (`prepare_time_s` in `summary.json`) to inference by scoring. Suppose a small adapted model with latency ℓ_{small} per decision matches a larger zero-shot model with latency ℓ_{large} . Its training time T_{train} is repaid after

$$n^* = \frac{T_{\text{train}}}{\ell_{\text{large}} - \ell_{\text{small}}}$$

decisions, [\[result pending: \[\$n^*\$ for the H3 pairs that match\]\]](#). The frontier in Figure 3 leaves this training cost out; including it moves [\[result pending: \[which cells drop off the frontier\]\]](#). For H7c we divide s by latency per decision. A negative s makes that ratio hard to read, so we compare ratios only between cells whose scores are positive [\[TODO: confirm this reading of the pre-registered H7c criterion before the tag\]](#).

6.3 What the feature probe controls for

The feature probe is a control, but what it controls for depends on what the feature vector ϕ contains, and the seven tasks differ in this respect.

On loan approval the default model in Equation (5) is a logistic function of the displayed fields, linear apart from the logarithmic loan-to-income term and the cap on years of employment, and the oracle thresholds it. A linear head on ϕ can therefore come close to the oracle given enough training states, and it sets a high bar: a language-model method that does not clear it adds nothing beyond the numbers it was shown. On the shifted variants the feature probe, like every adapted method, is trained on the unshifted task. Under the sign flip it has learned the wrong sign for the credit score and should fall far below its in-distribution score; under covariate shift the default model is unchanged, but the probe must extrapolate to lower incomes and credit scores. Its drop under each shift is the reference against which H3b’s drops for language models are read.

On the contextual bandit ϕ is the current context alone, and the weights w_a are redrawn for every episode. For any fixed function of the context, each option is then equally likely to be the best one, so the expected normalized score of the feature probe is exactly zero and the control measures nothing on this task. A language-model method that does better must be using the history in its prompt, which is the in-context learning the task exists to test.

On the bandit ϕ holds pull frequencies and empirical means, and the training labels come from a clairvoyant oracle. Under the behavior policy the oracle’s arm is also the most-pulled arm, so a cloned policy can learn to repeat whichever arm it has pulled most, which is pure exploitation ([Laskin et al., 2023](#); [Nie et al., 2025](#)). The same applies to probe and `lora_sft`, whose prompts show the same counts, and to the few-shot demonstrations, whose states come from oracle rollouts, so that the only arm ever pulled in them is the oracle’s. Low bandit scores for adapted methods therefore say more about cloning a clairvoyant teacher than about the model.

On gridworld, tic-tac-toe and blackjack the oracle is a non-linear function of ϕ : a shortest path around walls, a game-tree search, and a lookup table whose stand region is not monotone in the dealer’s card. A linear head on raw features is a weak control there. The gap between probe and `feature_probe` measures how much of the decision the model’s representation makes linearly decodable, which is not the same as the model using it ([Belinkov, 2022](#)). The gap between probe and `prompt_score` on the same model measures how much the model knows but does not show ([Orgad et al., 2025](#)), and the comparison of middle and last layers tests whether that information peaks before the output ([Skean et al., 2025](#)). [\[result pending: \[Probe minus feature probe and probe minus scoring, by task and size\]\]](#). The training statistics in

summary.json, probe training accuracy and LoRA loss before and after training, separate a method that failed to fit its training states from one that fitted them and failed to generalize [result pending: [cells with low training accuracy]].

6.4 Oracle agreement and return

Oracle agreement, the share of decisions in which the policy’s own legal action equals the logged oracle action, is easy to read but has two flaws, which is why return is the primary outcome. It is a lower bound on the share of optimal decisions whenever several actions are optimal, because the oracle breaks ties by a fixed rule. In tic-tac-toe against the random opponent the oracle does not prefer quicker wins: in the state shown in Appendix A.5, four cells win with certainty, including an immediate win at cell 7, and the oracle picks cell 5. On the two bandits it also measures agreement with a clairvoyant oracle that no learner can match early in an episode; UCB1 agrees with it in [result pending: [share]] of its decisions. Fallback actions after format failures never count as agreement. For each task we report the rank correlation between agreement and s across cells [result pending: [Kendall’s τ by task]] and look at the cells where the two disagree. High agreement with low return points to errors at decisive states, such as a missed block in tic-tac-toe; low agreement with high return points to alternative optimal actions, which the set-valued optimality of Section 6.1 resolves.

7 Limitations

7.0.0.1 Synthetic tasks. The seven tasks are small: at most ten actions, at most 100 decisions per episode, and states that fit in a few lines of text. We chose them because each has an exact oracle and a distinct decision structure, not because they resemble deployed agentic work, which involves long contexts, tool outputs and open-ended actions. Slopes measured here need not transfer to such settings, and a model’s rank on DEAD EYE is evidence about the decision structures we test, not about agentic ability in general. **7.0.0.2 Oracles.** Every oracle is optimal for its task, but several properties of the oracles limit what agreement and cloning can show. They break ties by fixed rules, so oracle agreement understates the share of optimal decisions wherever several moves are equally good. The blackjack oracle is optimal only for a reduced game without doubling, splitting or surrender, so our blackjack is not the casino game. The two bandit oracles are clairvoyant: a normalized score of 1 is out of reach there, and the labels used to train probes and LoRA come from a teacher whose choices cannot be inferred from the observation. A learnable teacher such as UCB1 would make behavior cloning on the bandits meaningful (Laskin *et al.*, 2023; Nie *et al.*, 2025); v1 does not use one. **7.0.0.3 Scale within families only.** We draw scale conclusions only within families, and even there parameter count is confounded with the number of training tokens per parameter (Hoffmann *et al.*, 2022; Kumar *et al.*, 2025). The largest rungs run with 4-bit weights, so the top of each ladder mixes scale with quantization, and the quantization ablation measures that cost only at 1.5B, 7B and 14B. Quantized rows report the catalog’s total parameter count rather than the packed weight count, and total counts overstate the capacity of the smallest models, in which embeddings take a large share. Comparisons across families are descriptive. **7.0.0.4 Prompt sensitivity.** Each condition uses one frozen prompt. Changes of format that preserve meaning can move accuracy by tens of points (Sclar *et al.*, 2024), and our format ablations, letters against names, demonstrations, temperature and length normalization, cover only a few such changes on one ladder. We do not measure sensitivity to wording, and a different prompt might move individual cells by more than the effects we test. **7.0.0.5 Scoring served models.** Models behind an OpenAI-compatible API are scored by the log-probability of the first token of each label, taken from the top 20 alternatives. This is exact only when the labels start with distinct tokens, which is why we use letters for such models, and letter labels bring option-identifier priors with them (Zheng *et al.*, 2024). Served models also apply their own sampling defaults. Their results are reported separately and never enter the scale curves. **7.0.0.6 Contamination.** Instances are generated from seeds and never appear on the web, but the rules and strategies of tic-tac-toe and blackjack are widely published, so part of a model’s performance there may be recall of strategy rather than reasoning. The contextual bandit and the sign-flipped loan task are the cases in which prior knowledge cannot help or actively misleads. **7.0.0.7 Adaptation.** Our fine-tuning is behavior cloning with LoRA, at one rank and one learning rate for every rung. Reinforcement-learning fine-tuning, which generalizes differently (Chu *et al.*, 2025) and repairs some exploration failures (Schmied *et al.*, 2026), is not part of v1, so our conclusions about small fine-tuned models concern cloning only. Out-of-distribution performance is tested on one task with two kinds of shift, which says nothing about shifts in task structure. **7.0.0.8 Statistics.** One hundred episodes per cell detect a difference of

0.10 with power 0.8 only when the standard deviation of the paired differences is at most about 0.36 (Section 4.5). On tasks whose anchors lie close together, blackjack above all, power is lower even with twenty hands per episode, and a null result there is weak evidence of no effect; we report the minimum detectable difference wherever it exceeds the effect of interest. The anchors themselves are means over the evaluation seeds and we treat them as fixed, so their sampling error is not propagated into the confidence intervals of s ; the paired tests do not depend on the anchors. **7.0.0.9 Decision models.** The comparison with Jev and its open-weight relatives is exploratory. The hosted model’s weights, architecture and training data are closed, so none of the controls we apply to our own ladders (same recipe, same training seeds, contamination checks on open data) apply to it, and its latency includes a network round trip. The open-weight decision models were trained on classification panels, not on our tasks, so a gap in either direction may reflect training data as much as method. **7.0.0.10 Hardware.** Latency depends on the hardware, the batch size and the implementation. Our cost comparisons hold for the hardware we report, and token counts are the better guide for other settings.

8 Conclusion

We built DEAD EYE to separate three things that evaluations of language-model agents usually run together: how large the model is, how it is turned into a policy, and what kind of decision it faces. [\[result pending: \[Two or three sentences on the main findings: the slope under generation and how it differs between planning and one-shot tasks; whether adaptation lets a small model match a ten times larger one, and where it breaks under shift; how much of the small-model deficit scoring recovers.\]\]](#) For someone choosing a model for a decision task, the practical reading is [\[result pending: \[recommendation on model size and conversion method, conditional on the findings\]\]](#). The next steps follow from the limitations: reinforcement-learning conversions, learnable teachers for the bandit tasks, and longer text environments in which the oracle is harder to compute.

8.0.0.1 Reproducibility statement. The harness, configurations, prompts and analysis code are released under the Apache-2.0 license at the repository linked on the title page, and the pre-registration is frozen at the git tag prereg-v1. Every result table and figure is produced by `deadeye report` from logged runs, and every number in the text is taken from the report’s `cells.csv` or `pairwise_methods.csv` or from `deadeye compare`. Appendix A reproduces every prompt verbatim, Appendix B the task generators, Appendix C the pre-registered criteria, Appendix D the compute budget and Appendix E the commands that rebuild the paper from a fresh clone. [\[TODO: Archive DOI for the code, results and report.\]](#)

9 Prompts

This appendix reproduces the messages a model receives, as printed by `deadeye show-prompt` for seed 0 after two oracle steps. The text is verbatim with two exceptions made for typesetting. Each system message is a single line in the prompt and is wrapped here at word boundaries, and the chain-of-thought instruction, also a single line, is wrapped the same way. Lines reading `[[user]]` and `[[assistant]]` mark the start of a chat message and are not part of the prompt. The system message is the same in every variant of a task, so it is shown once per task.

A few details are easier to state than to show. The probe embeds the generation-mode prompt, shown as the default variant, and LoRA trains and acts with the scoring-mode prompt. The prompts use each environment’s default parameters, which match the main sweep except for gridworld: the sweep’s grid is 6×6 , so its system message reads `6x6 grid and at most 24 moves and its move counter (max 24)`. For blackjack at seed 0 the hand ends with the second oracle action, a stand, so the state shown is the decision after one hit. In the tic-tac-toe state shown, cells 5, 6, 7 and 9 all win with certainty and the oracle picks cell 5, although cell 7 wins at once (Section 6.4). Few-shot demonstrations come from training seeds, so in tic-tac-toe they can show the agent playing the other mark, as the second demonstration below does; for the shifted loan variants they come from the unshifted task. When demonstrations are combined with chain-of-thought, each demonstration’s assistant turn carries the fixed rationale `Choosing the best available option.` before its action line. Base checkpoints receive the same messages without a chat template: the

system text, then each user turn after a line `Input:` and each assistant turn after a line `Output:`, separated by blank lines and ending with a line `Output:`.

9.1 Multi-armed bandit (bandit)

System message (identical in every variant; a single line in the prompt, wrapped here):

```
You are playing a 5-armed bandit for 50 rounds. Each round you choose one arm and receive a
reward of 1 or 0. Each arm has a fixed but unknown success probability. Your goal is to
maximise
the total reward over all rounds, which requires balancing exploration (trying arms to learn
about them) and exploitation (choosing the arm that looks best).
```

9.1.0.1 Default: free-form generation, action names. Command: `deadeye show-prompt bandit --seed 0 --steps 2`; oracle action in this state: `arm_5`.

```
[[user]]
Round 3 of 50.
Summary of past pulls:
  arm_1: never pulled
  arm_2: never pulled
  arm_3: never pulled
  arm_4: never pulled
  arm_5: pulled 2 times, average reward 0.00
Legal actions: arm_1, arm_2, arm_3, arm_4, arm_5
Reply with exactly one line of the form `Action: <name>` and nothing else.
```

9.1.0.2 Letter action format. Command: `deadeye show-prompt bandit --seed 0 --steps 2 --action-format letter`; oracle action in this state: `arm_5`.

```
[[user]]
Round 3 of 50.
Summary of past pulls:
  arm_1: never pulled
  arm_2: never pulled
  arm_3: never pulled
  arm_4: never pulled
  arm_5: pulled 2 times, average reward 0.00
Legal actions: A, B, C, D, E
  A = arm_1
  B = arm_2
  C = arm_3
  D = arm_4
  E = arm_5
Reply with exactly one line of the form `Action: <letter>` and nothing else.
```

9.1.0.3 Chain of thought. Command: `deadeye show-prompt bandit --seed 0 --steps 2 --cot`; oracle action in this state: `arm_5`.

```
[[user]]
Round 3 of 50.
Summary of past pulls:
  arm_1: never pulled
  arm_2: never pulled
  arm_3: never pulled
  arm_4: never pulled
  arm_5: pulled 2 times, average reward 0.00
Legal actions: arm_1, arm_2, arm_3, arm_4, arm_5
```

Think step by step in at most three short sentences, then end your reply with a final line of the form `Action: <name>`.

9.1.0.4 Likelihood scoring (also the LoRA training and inference prompt). Command: `deadeye show-prompt bandit --seed 0 --steps 2 --method prompt_score`; oracle action in this state: `arm_5`.

```
[[user]]
Round 3 of 50.
Summary of past pulls:
  arm_1: never pulled
  arm_2: never pulled
  arm_3: never pulled
  arm_4: never pulled
  arm_5: pulled 2 times, average reward 0.00
Legal actions: arm_1, arm_2, arm_3, arm_4, arm_5
Answer with only the action name of your chosen action.
```

9.1.0.5 Two oracle demonstrations from training seeds. Command: `deadeye show-prompt bandit --seed 0 --steps 2 --few-shot 2`; oracle action in this state: `arm_5`.

```
[[user]]
Round 4 of 50.
Summary of past pulls:
  arm_1: pulled 3 times, average reward 1.00
  arm_2: never pulled
  arm_3: never pulled
  arm_4: never pulled
  arm_5: never pulled
Legal actions: arm_1, arm_2, arm_3, arm_4, arm_5
Reply with exactly one line of the form `Action: <name>` and nothing else.
[[assistant]]
Action: arm_1
[[user]]
Round 2 of 50.
Summary of past pulls:
  arm_1: never pulled
  arm_2: never pulled
  arm_3: pulled 1 times, average reward 1.00
  arm_4: never pulled
  arm_5: never pulled
Legal actions: arm_1, arm_2, arm_3, arm_4, arm_5
Reply with exactly one line of the form `Action: <name>` and nothing else.
[[assistant]]
Action: arm_3
[[user]]
Round 3 of 50.
Summary of past pulls:
  arm_1: never pulled
  arm_2: never pulled
  arm_3: never pulled
  arm_4: never pulled
  arm_5: pulled 2 times, average reward 0.00
Legal actions: arm_1, arm_2, arm_3, arm_4, arm_5
Reply with exactly one line of the form `Action: <name>` and nothing else.
```

9.2 Contextual bandit (`contextual_bandit`)

System message (identical in every variant; a single line in the prompt, wrapped here):

You are making 30 sequential decisions. Each round you see a context made of 4 numeric features (x1..x4) and must choose one of 3 options (option_A, option_B, option_C). Each option's success probability depends on the context through a fixed but unknown rule. You receive a reward of 1 (success) or 0 (failure). Use the history of past contexts, choices and rewards to infer which option works best for which contexts, and maximise total reward.

9.2.0.1 Default: free-form generation, action names. Command: `deadeye show-prompt contextual_bandit --seed 0 --steps 2`; oracle action in this state: option_B.

```
[[user]]
Round 3 of 30.
Past rounds (most recent last, showing up to 10):
  context [x1=-2.33, x2=-0.22, x3=-1.25, x4=-0.73] -> chose option_C -> reward 1
  context [x1=-0.54, x2=-0.32, x3=+0.41, x4=+1.04] -> chose option_B -> reward 1
Current context: [x1=-0.13, x2=+1.37, x3=-0.67, x4=+0.35]
Legal actions: option_A, option_B, option_C
Reply with exactly one line of the form `Action: <name>` and nothing else.
```

9.2.0.2 Letter action format. Command: `deadeye show-prompt contextual_bandit --seed 0 --steps 2 --action-format letter`; oracle action in this state: option_B.

```
[[user]]
Round 3 of 30.
Past rounds (most recent last, showing up to 10):
  context [x1=-2.33, x2=-0.22, x3=-1.25, x4=-0.73] -> chose option_C -> reward 1
  context [x1=-0.54, x2=-0.32, x3=+0.41, x4=+1.04] -> chose option_B -> reward 1
Current context: [x1=-0.13, x2=+1.37, x3=-0.67, x4=+0.35]
Legal actions: A, B, C
  A = option_A
  B = option_B
  C = option_C
Reply with exactly one line of the form `Action: <letter>` and nothing else.
```

9.2.0.3 Chain of thought. Command: `deadeye show-prompt contextual_bandit --seed 0 --steps 2 --cot`; oracle action in this state: option_B.

```
[[user]]
Round 3 of 30.
Past rounds (most recent last, showing up to 10):
  context [x1=-2.33, x2=-0.22, x3=-1.25, x4=-0.73] -> chose option_C -> reward 1
  context [x1=-0.54, x2=-0.32, x3=+0.41, x4=+1.04] -> chose option_B -> reward 1
Current context: [x1=-0.13, x2=+1.37, x3=-0.67, x4=+0.35]
Legal actions: option_A, option_B, option_C
Think step by step in at most three short sentences, then end your reply with a final line of the form `Action: <name>`.
```

9.2.0.4 Likelihood scoring (also the LoRA training and inference prompt). Command: `deadeye show-prompt contextual_bandit --seed 0 --steps 2 --method prompt_score`; oracle action in this state: option_B.

```
[[user]]
Round 3 of 30.
Past rounds (most recent last, showing up to 10):
  context [x1=-2.33, x2=-0.22, x3=-1.25, x4=-0.73] -> chose option_C -> reward 1
  context [x1=-0.54, x2=-0.32, x3=+0.41, x4=+1.04] -> chose option_B -> reward 1
```

```
Current context: [x1=-0.13, x2=+1.37, x3=-0.67, x4=+0.35]
Legal actions: option_A, option_B, option_C
Answer with only the action name of your chosen action.
```

9.2.0.5 Two oracle demonstrations from training seeds. Command: `deadeye show-prompt contextual_bandit -seed 0 --steps 2 --few-shot 2`; oracle action in this state: `option_B`.

```
[[user]]
Round 4 of 30.
Past rounds (most recent last, showing up to 10):
  context [x1=+0.25, x2=-1.16, x3=-0.21, x4=-0.16] -> chose option_A -> reward 1
  context [x1=-0.15, x2=+1.38, x3=-0.34, x4=-0.48] -> chose option_B -> reward 1
  context [x1=+0.63, x2=+0.54, x3=+0.61, x4=+1.11] -> chose option_C -> reward 1
Current context: [x1=+1.48, x2=+0.03, x3=-1.14, x4=-0.58]
Legal actions: option_A, option_B, option_C
Reply with exactly one line of the form `Action: <name>` and nothing else.
[[assistant]]
Action: option_B
[[user]]
Round 2 of 30.
Past rounds (most recent last, showing up to 10):
  context [x1=-0.82, x2=+0.71, x3=-0.75, x4=+0.41] -> chose option_A -> reward 1
Current context: [x1=-0.10, x2=-0.93, x3=+2.11, x4=+0.72]
Legal actions: option_A, option_B, option_C
Reply with exactly one line of the form `Action: <name>` and nothing else.
[[assistant]]
Action: option_C
[[user]]
Round 3 of 30.
Past rounds (most recent last, showing up to 10):
  context [x1=-2.33, x2=-0.22, x3=-1.25, x4=-0.73] -> chose option_C -> reward 1
  context [x1=-0.54, x2=-0.32, x3=+0.41, x4=+1.04] -> chose option_B -> reward 1
Current context: [x1=-0.13, x2=+1.37, x3=-0.67, x4=+0.35]
Legal actions: option_A, option_B, option_C
Reply with exactly one line of the form `Action: <name>` and nothing else.
```

9.3 Loan approval (loan)

System message (identical in every variant; a single line in the prompt, wrapped here):

```
You are a loan officer deciding whether to approve loan applications. For each applicant you
see
their profile and the loan terms. If you approve and the loan is repaid, the bank earns the
interest shown; if you approve and the applicant defaults, the bank loses the amount shown.
Denying earns and loses nothing. Approve an application only when you expect it to be
profitable
on average, using the applicant's creditworthiness. Your goal is to maximise total profit.
```

9.3.0.1 Default: free-form generation, action names. Command: `deadeye show-prompt loan --seed 0 --steps 2`; oracle action in this state: `approve`.

```
[[user]]
Applicant 3 of 20.
  Age: 41
  Annual income: $67.6k
  Existing debt-to-income ratio: 0.43
  Credit score: 752 (range 300-850)
  Years in current employment: 2
```

```
Requested loan: $9.6k for car
If repaid the bank earns $1.2k; if the applicant defaults the bank loses $5.8k.
Legal actions: approve, deny
Reply with exactly one line of the form `Action: <name>` and nothing else.
```

9.3.0.2 Letter action format. Command: `deadeye show-prompt loan --seed 0 --steps 2 --action-format letter`; oracle action in this state: approve.

```
[[user]]
Applicant 3 of 20.
Age: 41
Annual income: $67.6k
Existing debt-to-income ratio: 0.43
Credit score: 752 (range 300-850)
Years in current employment: 2
Requested loan: $9.6k for car
If repaid the bank earns $1.2k; if the applicant defaults the bank loses $5.8k.
Legal actions: A, B
A = approve
B = deny
Reply with exactly one line of the form `Action: <letter>` and nothing else.
```

9.3.0.3 Chain of thought. Command: `deadeye show-prompt loan --seed 0 --steps 2 --cot`; oracle action in this state: approve.

```
[[user]]
Applicant 3 of 20.
Age: 41
Annual income: $67.6k
Existing debt-to-income ratio: 0.43
Credit score: 752 (range 300-850)
Years in current employment: 2
Requested loan: $9.6k for car
If repaid the bank earns $1.2k; if the applicant defaults the bank loses $5.8k.
Legal actions: approve, deny
Think step by step in at most three short sentences, then end your reply with a final line of
the form `Action: <name>`.
```

9.3.0.4 Likelihood scoring (also the LoRA training and inference prompt). Command: `deadeye show-prompt loan --seed 0 --steps 2 --method prompt_score`; oracle action in this state: approve.

```
[[user]]
Applicant 3 of 20.
Age: 41
Annual income: $67.6k
Existing debt-to-income ratio: 0.43
Credit score: 752 (range 300-850)
Years in current employment: 2
Requested loan: $9.6k for car
If repaid the bank earns $1.2k; if the applicant defaults the bank loses $5.8k.
Legal actions: approve, deny
Answer with only the action name of your chosen action.
```

9.3.0.5 Two oracle demonstrations from training seeds. Command: `deadeye show-prompt loan --seed 0 --steps 2 --few-shot 2`; oracle action in this state: approve.

```
[[user]]
Applicant 4 of 20.
```

```

Age: 54
Annual income: $41.1k
Existing debt-to-income ratio: 0.34
Credit score: 727 (range 300-850)
Years in current employment: 0
Requested loan: $6.8k for debt consolidation
If repaid the bank earns $0.8k; if the applicant defaults the bank loses $4.1k.
Legal actions: approve, deny
Reply with exactly one line of the form `Action: <name>` and nothing else.
[[assistant]]
Action: approve
[[user]]
Applicant 2 of 20.
Age: 34
Annual income: $40.3k
Existing debt-to-income ratio: 0.20
Credit score: 686 (range 300-850)
Years in current employment: 23
Requested loan: $8.6k for small business
If repaid the bank earns $1.0k; if the applicant defaults the bank loses $5.2k.
Legal actions: approve, deny
Reply with exactly one line of the form `Action: <name>` and nothing else.
[[assistant]]
Action: approve
[[user]]
Applicant 3 of 20.
Age: 41
Annual income: $67.6k
Existing debt-to-income ratio: 0.43
Credit score: 752 (range 300-850)
Years in current employment: 2
Requested loan: $9.6k for car
If repaid the bank earns $1.2k; if the applicant defaults the bank loses $5.8k.
Legal actions: approve, deny
Reply with exactly one line of the form `Action: <name>` and nothing else.

```

9.4 Gridworld navigation (gridworld)

System message (identical in every variant; a single line in the prompt, wrapped here):

```

You are navigating a 5x5 grid. 'A' marks your position, 'G' the goal, '#' walls and '.' free
cells. Rows are numbered top to bottom and columns left to right, both starting at 0. Each
move
goes one cell up, down, left or right; you cannot move into walls or off the grid. Reach the
goal in as few moves as possible (at most 20 moves).

```

9.4.0.1 Default: free-form generation, action names. Command: `deadeye show-prompt gridworld --seed 0 --steps 2`; oracle action in this state: left.

```

[[user]]
Move 3 (max 20).
Map:
.#...
#.G..
...#.
...A#
#....
You are at row 3, column 3. The goal is at row 1, column 2.

```

```
Legal actions: down, left
Reply with exactly one line of the form `Action: <name>` and nothing else.
```

9.4.0.2 Letter action format. Command: `deadeye show-prompt gridworld --seed 0 --steps 2 --action-format letter`; oracle action in this state: left.

```
[[user]]
Move 3 (max 20).
Map:
.#...
#.G..
...#.
...A#
#....
You are at row 3, column 3. The goal is at row 1, column 2.
Legal actions: A, B
  A = down
  B = left
Reply with exactly one line of the form `Action: <letter>` and nothing else.
```

9.4.0.3 Chain of thought. Command: `deadeye show-prompt gridworld --seed 0 --steps 2 --cot`; oracle action in this state: left.

```
[[user]]
Move 3 (max 20).
Map:
.#...
#.G..
...#.
...A#
#....
You are at row 3, column 3. The goal is at row 1, column 2.
Legal actions: down, left
Think step by step in at most three short sentences, then end your reply with a final line of
the form `Action: <name>`.
```

9.4.0.4 Likelihood scoring (also the LoRA training and inference prompt). Command: `deadeye show-prompt gridworld --seed 0 --steps 2 --method prompt_score`; oracle action in this state: left.

```
[[user]]
Move 3 (max 20).
Map:
.#...
#.G..
...#.
...A#
#....
You are at row 3, column 3. The goal is at row 1, column 2.
Legal actions: down, left
Answer with only the action name of your chosen action.
```

9.4.0.5 Two oracle demonstrations from training seeds. Command: `deadeye show-prompt gridworld --seed 0 --steps 2 --few-shot 2`; oracle action in this state: left.

```
[[user]]
Move 4 (max 20).
Map:
.....
```

```

...A.
#.#G
.....
.#.#.
You are at row 1, column 3. The goal is at row 2, column 4.
Legal actions: up, left, right
Reply with exactly one line of the form `Action: <name>` and nothing else.
[[assistant]]
Action: right
[[user]]
Move 2 (max 20).
Map:
.A...
#.#..
..#..
..#.G
..#..
You are at row 0, column 1. The goal is at row 3, column 4.
Legal actions: down, left, right
Reply with exactly one line of the form `Action: <name>` and nothing else.
[[assistant]]
Action: right
[[user]]
Move 3 (max 20).
Map:
.#...
#.G..
...#.
...A#
#....
You are at row 3, column 3. The goal is at row 1, column 2.
Legal actions: down, left
Reply with exactly one line of the form `Action: <name>` and nothing else.

```

9.5 Tic-tac-toe (tictactoe)

System message (identical in every variant; a single line in the prompt, wrapped here):

```

You are playing tic-tac-toe as 'X' against an opponent playing 'O'. The board cells are
numbered
1 to 9 from left to right, top to bottom. Three of your marks in a row, column or diagonal
wins.
Choose the number of an empty cell. Win if you can, otherwise force a draw; never lose if
avoidable.

```

9.5.0.1 Default: free-form generation, action names. Command: `deadeye show-prompt tictactoe --seed 0 --steps 2`; oracle action in this state: 5.

```

[[user]]
You are 'X'. Board (numbers mark empty cells):
X 2 0
X 5 6
7 0 9
Legal actions: 2, 5, 6, 7, 9
Reply with exactly one line of the form `Action: <name>` and nothing else.

```

9.5.0.2 Letter action format. Command: `deadeye show-prompt tictactoe --seed 0 --steps 2 --action-format letter`; oracle action in this state: 5.

```

[[user]]
You are 'X'. Board (numbers mark empty cells):
X 2 0
X 5 6
7 0 9
Legal actions: A, B, C, D, E
  A = 2
  B = 5
  C = 6
  D = 7
  E = 9
Reply with exactly one line of the form `Action: <letter>` and nothing else.

```

9.5.0.3 Chain of thought. Command: `deadeye show-prompt tictactoe --seed 0 --steps 2 --cot; oracle`
 action in this state: 5.

```

[[user]]
You are 'X'. Board (numbers mark empty cells):
X 2 0
X 5 6
7 0 9
Legal actions: 2, 5, 6, 7, 9
Think step by step in at most three short sentences, then end your reply with a final line of
the form `Action: <name>`.

```

9.5.0.4 Likelihood scoring (also the LoRA training and inference prompt). Command: `deadeye show-prompt`
`tictactoe --seed 0 --steps 2 --method prompt_score; oracle` action in this state: 5.

```

[[user]]
You are 'X'. Board (numbers mark empty cells):
X 2 0
X 5 6
7 0 9
Legal actions: 2, 5, 6, 7, 9
Answer with only the action name of your chosen action.

```

9.5.0.5 Two oracle demonstrations from training seeds. Command: `deadeye show-prompt tictactoe --seed 0`
`--steps 2 --few-shot 2; oracle` action in this state: 5.

```

[[user]]
You are 'X'. Board (numbers mark empty cells):
1 2 3
4 5 6
7 8 9
Legal actions: 1, 2, 3, 4, 5, 6, 7, 8, 9
Reply with exactly one line of the form `Action: <name>` and nothing else.
[[assistant]]
Action: 1
[[user]]
You are '0'. Board (numbers mark empty cells):
1 2 3
4 0 6
X 8 X
Legal actions: 1, 2, 3, 4, 6, 8
Reply with exactly one line of the form `Action: <name>` and nothing else.
[[assistant]]
Action: 8
[[user]]
You are 'X'. Board (numbers mark empty cells):

```

```
X 2 0
X 5 6
7 0 9
Legal actions: 2, 5, 6, 7, 9
Reply with exactly one line of the form `Action: <name>` and nothing else.
```

9.6 Blackjack (blackjack)

System message (identical in every variant; a single line in the prompt, wrapped here):

```
You are playing blackjack against the dealer. Cards 2-10 are worth their number, face cards
are
worth 10 and an ace is worth 11 if that does not bust you, otherwise 1. You may 'hit' (take
another card) or 'stand' (end your turn). If you exceed 21 you bust and lose. After you
stand,
the dealer draws until reaching 17 or more. You win if your total is higher than the dealer's
or
the dealer busts; equal totals tie. A two-card 21 pays extra. Maximise your expected
winnings.
```

9.6.0.1 Default: free-form generation, action names. Command: `deadeye show-prompt blackjack --seed 0 --steps 2`; oracle action in this state: stand.

```
[[user]]
Your cards: 7, 2, 10 (total 19).
Dealer shows: 3.
Legal actions: hit, stand
Reply with exactly one line of the form `Action: <name>` and nothing else.
```

9.6.0.2 Letter action format. Command: `deadeye show-prompt blackjack --seed 0 --steps 2 --action-format letter`; oracle action in this state: stand.

```
[[user]]
Your cards: 7, 2, 10 (total 19).
Dealer shows: 3.
Legal actions: A, B
  A = hit
  B = stand
Reply with exactly one line of the form `Action: <letter>` and nothing else.
```

9.6.0.3 Chain of thought. Command: `deadeye show-prompt blackjack --seed 0 --steps 2 --cot`; oracle action in this state: stand.

```
[[user]]
Your cards: 7, 2, 10 (total 19).
Dealer shows: 3.
Legal actions: hit, stand
Think step by step in at most three short sentences, then end your reply with a final line of
the form `Action: <name>`.
```

9.6.0.4 Likelihood scoring (also the LoRA training and inference prompt). Command: `deadeye show-prompt blackjack --seed 0 --steps 2 --method prompt_score`; oracle action in this state: stand.

```
[[user]]
Your cards: 7, 2, 10 (total 19).
Dealer shows: 3.
```

```
Legal actions: hit, stand
Answer with only the action name of your chosen action.
```

9.6.0.5 Two oracle demonstrations from training seeds. Command: `deadeye show-prompt blackjack --seed 0 --steps 2 --few-shot 2`; oracle action in this state: `stand`.

```
[[user]]
Your cards: 8, 5 (total 13).
Dealer shows: 2.
Legal actions: hit, stand
Reply with exactly one line of the form `Action: <name>` and nothing else.
[[assistant]]
Action: stand
[[user]]
Your cards: 6, 10 (total 16).
Dealer shows: 7.
Legal actions: hit, stand
Reply with exactly one line of the form `Action: <name>` and nothing else.
[[assistant]]
Action: hit
[[user]]
Your cards: 7, 2, 10 (total 19).
Dealer shows: 3.
Legal actions: hit, stand
Reply with exactly one line of the form `Action: <name>` and nothing else.
```

9.7 Customer-support triage (support)

System message (identical in every variant; the policy text is part of it):

```
You are a customer support agent deciding how to resolve tickets. For each ticket you see the
customer's message and the order facts. Apply the company policy exactly:
1. If the customer has 3 or more prior claims and the issue is a damaged item, a wrong item
or a
missing delivery, escalate to the fraud team.
2. Billing errors: refund.
3. Changed mind: refund if the item was delivered at most 14 days ago (30 days for premium
customers), otherwise deny.
4. Late delivery (item received): partial_refund if it was more than 7 days late, otherwise
deny.
5. Not received: if it is at most 3 days past the promised date, deny (ask the customer to
wait); if later and the order value is over $100, escalate; otherwise refund.
6. Damaged or wrong item: if no photo evidence is attached and the order value is over $50,
request_evidence; otherwise refund if the customer asks for a refund, else replace if a
replacement is in stock, else refund.
Correct decisions earn credit; unwarranted payouts cost the order value and refusing a
legitimate claim loses the customer.
```

9.7.0.1 Default: free-form generation, action names. Command: `deadeye show-prompt support --seed 0 --steps 2`; oracle action in this state: `escalate`.

```
[[user]]
Ticket 3 of 10.
Customer message: "Wrong item in the box. I don't want it, please refund."
Order: backpack, value $59.79. Delivered 53 days ago.
Photo evidence attached: no. Prior claims by this customer: 3. Customer tier: standard.
Replacement in stock: yes.
```

Legal actions: refund, replace, partial_refund, request_evidence, escalate, deny
Reply with exactly one line of the form `Action: <name>` and nothing else.

9.7.0.2 Free reply (the unconverted assistant). Command: deadeye show-prompt support --seed 0 --steps 2 --method prompt_free; oracle action in this state: escalate.

```
[[user]]
Ticket 3 of 10.
Customer message: "Wrong item in the box. I don't want it, please refund."
Order: backpack, value $59.79. Delivered 53 days ago.
Photo evidence attached: no. Prior claims by this customer: 3. Customer tier: standard.
  Replacement in stock: yes.
Legal actions: refund, replace, partial_refund, request_evidence, escalate, deny
What do you do? Answer in your own words.
```

9.7.0.3 Letter action format. Command: deadeye show-prompt support --seed 0 --steps 2 --action-format letter; oracle action in this state: escalate.

```
[[user]]
Ticket 3 of 10.
Customer message: "Wrong item in the box. I don't want it, please refund."
Order: backpack, value $59.79. Delivered 53 days ago.
Photo evidence attached: no. Prior claims by this customer: 3. Customer tier: standard.
  Replacement in stock: yes.
Legal actions: A, B, C, D, E, F
  A = refund
  B = replace
  C = partial_refund
  D = request_evidence
  E = escalate
  F = deny
Reply with exactly one line of the form `Action: <letter>` and nothing else.
```

9.7.0.4 Chain of thought. Command: deadeye show-prompt support --seed 0 --steps 2 --cot; oracle action in this state: escalate.

```
[[user]]
Ticket 3 of 10.
Customer message: "Wrong item in the box. I don't want it, please refund."
Order: backpack, value $59.79. Delivered 53 days ago.
Photo evidence attached: no. Prior claims by this customer: 3. Customer tier: standard.
  Replacement in stock: yes.
Legal actions: refund, replace, partial_refund, request_evidence, escalate, deny
Think step by step in at most three short sentences, then end your reply with a final line of
the form `Action: <name>`.
```

9.7.0.5 Likelihood scoring (also the LoRA training and inference prompt). Command: deadeye show-prompt support --seed 0 --steps 2 --method prompt_score; oracle action in this state: escalate.

```
[[user]]
Ticket 3 of 10.
Customer message: "Wrong item in the box. I don't want it, please refund."
Order: backpack, value $59.79. Delivered 53 days ago.
Photo evidence attached: no. Prior claims by this customer: 3. Customer tier: standard.
  Replacement in stock: yes.
Legal actions: refund, replace, partial_refund, request_evidence, escalate, deny
Answer with only the action name of your chosen action.
```

9.7.0.6 Two oracle demonstrations from training seeds. Command: `deadeye show-prompt support --seed 0 --steps 2 --few-shot 2`; oracle action in this state: `escalate`.

```
[[user]]
Ticket 4 of 10.
Customer message: "I don't want the backpack anymore. It's unused."
Order: backpack, value $46.76. Delivered 54 days ago.
Photo evidence attached: yes. Prior claims by this customer: 1. Customer tier: premium.
  Replacement in stock: yes.
Legal actions: refund, replace, partial_refund, request_evidence, escalate, deny
Reply with exactly one line of the form `Action: <name>` and nothing else.
[[assistant]]
Action: deny
[[user]]
Ticket 2 of 10.
Customer message: "There's a wrong charge on my phone case order, please fix it."
Order: phone case, value $67.26. Delivered 8 days ago.
Photo evidence attached: no. Prior claims by this customer: 2. Customer tier: standard.
  Replacement in stock: yes.
Legal actions: refund, replace, partial_refund, request_evidence, escalate, deny
Reply with exactly one line of the form `Action: <name>` and nothing else.
[[assistant]]
Action: refund
[[user]]
Ticket 3 of 10.
Customer message: "Wrong item in the box. I don't want it, please refund."
Order: backpack, value $59.79. Delivered 53 days ago.
Photo evidence attached: no. Prior claims by this customer: 3. Customer tier: standard.
  Replacement in stock: yes.
Legal actions: refund, replace, partial_refund, request_evidence, escalate, deny
Reply with exactly one line of the form `Action: <name>` and nothing else.
```

10 Task parameters and generators

Table 8 lists every task parameter with its default, its value in the main sweep and its values in the ablations. The rest of this appendix gives the generators in full.

Task parameters. Ablation labels are the task-configuration keys used in the five files under `configs/ablations/`; blank cells take the main-sweep value.

Task	Parameter	Default	Main sweep	Ablation variants
bandit	n_arms	5	5	10 (bandit_hard)
	horizon	50	50	100 (bandit_hard)
	gap	0.2	0.2	
	means_mode	gap	gap	
	history_format	summary	summary	raw (bandit_rawhist)
contextual_bandit	n_actions	3	3	
	dim	4	4	
	horizon	30	30	
	history_window	10	10	
	weight_scale	2.0	2.0	
loan	n_applicants	20	20	

Task	Parameter	Default	Main sweep	Ablation variants
	interest_rate	0.12	0.12	
	loss_given_default	0.6	0.6	
	shift	none	none	covariate(loan_covariate_shift), sign_flip(loan_sign_flip); both trained on none
gridworld	size	5	6	9 (gridworld_large)
	n_walls	size	6	16
	max_steps	4×size	24	36
	step_penalty	0.02	0.02	
	min_distance	size	6	9
tictactoe	opponent	random	random	minimax(tictactoe_minimax)
	agent_mark	alternate	alternate	
blackjack	natural_bonus	true	true	
	hands_per_episode	1	20	20
	episodes		100	100
support	n_tickets	10	20	40 (support_long)

10.0.0.1 Bandit. With means_mode set to gap, the index of the best arm is drawn uniformly and the reward table, T rows by K columns of Bernoulli draws, follows from the same generator; the alternative uniform mode, which draws each μ_a from $U(0.1, 0.9)$, is not used in our configurations. The pseudo-regret is computed from the true means, not from the realized rewards. The feature vector has $2K$ entries: each arm’s share of the pulls so far and its empirical mean, with 0.5 for unpulled arms. An episode has exactly T decisions.

10.0.0.2 Contextual bandit. The generator draws the weight matrix ($A \times d$, entries from $\mathcal{N}(0, 2^2)$), then the $T \times d$ contexts (entries from $\mathcal{N}(0, 1)$, rounded to two decimals), then the $T \times A$ table of Bernoulli outcomes. Regret uses the true success probabilities. The feature vector is the current context. An episode has exactly T decisions.

10.0.0.3 Loan approval. For each applicant the generator draws the annual income in thousands of dollars, $I = \text{clip}(e^Z, 12, 400)$ with $Z \sim \mathcal{N}(\ln 55, 0.5^2)$ and rounded to one decimal; the credit score, $C = \lfloor \text{clip}(\mathcal{N}(680, 70^2), 300, 850) \rfloor$; the age, uniform on $\{21, \dots, 70\}$; the debt-to-income ratio, $D = \text{clip}(\text{Beta}(2, 5), 0, 0.9)$ rounded to two decimals; the years in current employment, $E = \lfloor \text{clip}(\text{Exp}(\text{mean } 6), 0, 35) \rfloor$; the requested amount in thousands, $L = \text{clip}(e^{Z'}, 1, 150)$ with $Z' \sim \mathcal{N}(\ln 15, 0.6^2)$ and rounded to one decimal; and a purpose drawn uniformly from car, home improvement, small business, education and debt consolidation, with offsets $\rho = -0.2, -0.3, +0.6, 0$ and $+0.4$ in Equation (5). Defaults are then drawn as Bernoulli(p_i) for all applicants at once. Covariate shift replaces $\ln 55$ by $\ln 35$ and the credit mean 680 by 600 and leaves the default model unchanged; the sign flip sets $c = +1$ and leaves the applicant distribution unchanged, so a sign-flipped seed shows the same applicants as the unshifted one. The feature vector holds age, income, debt ratio, credit score, years employed and amount, standardized as $(x - \mu)/s$ with (μ, s) equal to $(45, 15)$, $(55, 40)$, $(0.28, 0.15)$, $(680, 70)$, $(6, 6)$ and $(15, 15)$, followed by a one-hot purpose. An episode has exactly N decisions.

10.0.0.4 Gridworld. The generator permutes the n^2 cells; the first becomes the start, the second the goal and the next n_{walls} become walls. It redraws until breadth-first search from the goal reaches the start at distance $d^* \geq \text{min_distance}$. The feature vector holds the agent’s and the goal’s row and column divided by n , followed by the n^2 wall indicators. An episode has between d^* and $4n$ decisions.

10.0.0.5 Tic-tac-toe. The agent plays X on even seeds and moves first, and O on odd seeds, in which case the opponent opens. Both opponents draw from a generator seeded with $\sigma + 10007$: the random opponent chooses uniformly among empty cells, and the minimax opponent chooses uniformly among the moves of highest depth-aware minimax value from its own point of view. The feature vector holds the nine cells as $+1$ for the agent’s marks, -1 for the opponent’s and 0 for empty ones. An episode has between two and five decisions.

10.0.0.6 Blackjack. A generator seeded with $\sigma + 20011$ draws two

streams of 32 cards, one for the player and one for the dealer, each card uniform over the 13 entries $\{1, \dots, 9, 10, 10, 10, 10\}$. The player receives the first two cards of the player stream and the dealer the first two of the dealer stream, and the dealer’s first card is shown. The dealer’s hand is not checked for a natural before the player acts, and a player’s two-card 21 against a dealer total of 21 is a tie. The feature vector is the hand total divided by 21, the soft-hand flag and the dealer’s card divided by 10. An episode has at least one decision and is capped at 12. **10.0.0.7 Customer-support triage.** A generator seeded with $\sigma + 30013$ draws, for each of the n tickets, the issue type uniformly from six (damaged, not received, wrong item, late, changed mind, billing error), what the customer asks for (refund with probability 0.45, replacement 0.25, nothing specific 0.30), the item from a list of eight, the order value $\text{clip}(e^Z, 5, 500)$ with $Z \sim \mathcal{N}(\ln 60, 0.8^2)$, the days since delivery (uniform on 0 to 60), the days past the promised date (uniform on 0 to 14), whether photo evidence is attached (probability 0.5), the number of prior claims (from $\{0, 0, 0, 1, 1, 2, 3, 4\}$), the customer tier (premium with probability 0.3) and whether a replacement is in stock (probability 0.7). The customer message is one of two or three fixed sentences per issue and request, with the item name filled in; one of them is the familiar “I received the package damaged. I need a refund today.” The six-rule policy in the task description is the oracle. A correct decision earns $+1$; a wrong one -1 , less the payout in hundreds of dollars when the agent refunds, replaces (or half of it for a partial refund) where the policy said deny, request evidence or escalate, and less a further 1 when it denies a claim the policy would have paid. The feature vector is the one-hot issue and request, the scaled timing, value, evidence, prior-claim, tier and stock fields. **10.0.0.8 Other sources of randomness.** The random baseline uses one generator per cell, seeded with 0. When a decision is illegal, the substitute action comes from a generator seeded with $\sigma + 7$ for that episode. The behavior policy that collects probe and LoRA training states uses one generator, seeded with 0, across all training episodes. Few-shot demonstrations take one state from each training seed in order, starting at 100000; a generator seeded with 12345 draws the number of oracle steps taken before the demonstration state uniformly from 0 to $\min(5, H) - 1$, and if the episode ends during those steps the initial state is used instead. Greedy decoding uses no randomness; for sampled decoding the model’s generator is seeded with the episode seed before each episode.

11 Pre-registered predictions and falsification criteria

This appendix restates docs/preregistration.md as frozen at the tag prereg-v1. **[TODO: Re-copy after the tag is created, and record any deviation below.]**

11.0.0.1 Research questions. RQ1: how does decision quality scale with parameter count within a model family, and does the slope depend on the task’s decision structure? RQ2: does the conversion method (prompting, likelihood scoring, hidden-state probe, LoRA behavior cloning) change the dependence on scale? RQ3: which other factors (instruction tuning, quantization, reasoning budget, observation format, prompt encoding, adaptation data budget) move decision quality, and by how much? RQ4: what is the frontier of quality against inference cost? **11.0.0.2 Predictions.** Table 9 lists every prediction with its criteria. “Score” is the normalized score s ; “CI” is a 95% bootstrap confidence interval; slopes are the b of Equation (9), fitted within one family on the sweep results.

Pre-registered predictions with the conditions under which each is supported or falsified. Outcomes between the two conditions are reported as inconclusive.

	Prediction	Supported if	Falsified if
H1a	Under <code>prompt_generate</code> , $b > 0$ in every family	the CI of b excludes 0 and is positive for every family with at least four sizes	any family’s CI includes 0 or is negative
H1b	The slope is larger for gridworld and tic-tac-toe than for loan	the bootstrap CI of $b_{\text{plan}} - b_{\text{loan}}$ excludes 0	that CI includes 0
H2a	Instruct beats base under <code>prompt_generate</code> at every size	the CI of the paired difference lies above 0 at 80% of sizes or more	it does so at fewer than 50% of sizes
H2b	The instruct-minus-base gap under <code>probe</code> and <code>lora_sft</code> is below 0.05	the CI of the mean gap lies within $[-0.05, 0.05]$	the CI excludes that interval

	Prediction	Supported if	Falsified if
H3a	The best <code>lora_sft</code> model of at most 1.7B is at least as good as a <code>prompt_generate</code> model ten times larger, in distribution	the CI of the paired difference is at or above 0 on at least 4 of 6 tasks	this holds on fewer than 3 tasks
H3b	On <code>loan_sign_flip</code> and <code>loan_covariate_shift</code> , the small adapted model loses more relative to in distribution than the large model	the CI of the interaction excludes 0	that CI includes 0
H4a	The illegal-action rate under <code>prompt_score</code> is 0	by construction	not applicable
H4b	$b_{\text{score}} < 0.5 b_{\text{gen}}$	the bootstrap CI of the ratio lies below 0.5	the CI includes 0.5 or lies above it
H5	int4 costs less than 0.05 at 7B and above and more than 0.10 below 2B	the CIs of the paired bf16-minus-int4 differences meet both thresholds	either threshold fails
H6a	No model under prompting reaches the UCB1 score on the bandit	every CI lies below the UCB1 score	any CI includes or exceeds it
H6b	Summary history beats raw history for models below 2B	the paired CI lies above 0 for every such model	any such CI includes 0
H7a	Thinking or chain-of-thought improves gridworld and tic-tac-toe at 4B and above	the paired CI lies above 0	the CI includes 0
H7b	Thinking or chain-of-thought does not improve models below 2B	the paired CI includes 0 or is negative	the CI lies above 0
H7c	Latency-normalized score is lower with thinking at every size	the ratio of score to latency is lower at every size	it is higher at any size

11.0.0.3 Design, analysis and exclusions. Tasks, parameters and seeds are those of the configuration files at the tag; evaluation seeds are 0 to 99 (0 to 299 for tic-tac-toe; blackjack plays twenty hands per episode) and training seeds 100000 to 100099, and prompts are frozen at the tag. The primary outcome is the normalized return per episode; secondary outcomes are the illegal-action rate, oracle agreement, the task metrics, and latency and tokens per decision. Each cell is summarized by its mean, interquartile mean and percentile bootstrap interval (2000 resamples); paired comparisons use sign-flip permutation tests (5000 permutations) with Holm’s correction within each hypothesis’s family of tests; slopes are fitted by ordinary least squares with bootstrap intervals over episodes. No episode is excluded because of its outcome. A cell is excluded only if its model fails to load or its run crashes, as recorded in `run_manifest.json`, and no interim analysis changes the design. **11.0.0.4 Deviations and excluded cells.** [TODO: Copy the deviations table of `docs/preregistration.md` (date, change, reason) and list every excluded cell with the reason recorded in `run_manifest.json`; write “none” if there are none.]

12 Compute budget

The figures in this appendix are planning estimates from `docs/compute_budget.md` and from `deadeye_estimate`, not measurements. [TODO: Replace them with measured GPU-hours per configuration, summed from `eval_time_s` and `prepare_time_s` in `summary.json`.]

Planning estimates of seconds per decision, for prompts of 300 to 600 tokens and answers of 1 to 16 tokens.

Setting	<code>prompt_score</code>	<code>prompt_generate</code> (16 tokens)	probe embedding
135M to 360M, CPU (4 cores)	0.15 to 0.4	0.3 to 0.8	0.1 to 0.3
1.5B, CPU	1 to 2	2 to 4	0.8 to 1.5
1.5B, 24 GB GPU, bf16	0.03	0.08	0.02
7B, 24 GB GPU, bf16	0.08	0.25	0.06
14B, 24 GB GPU, bf16	0.15	0.5	0.12
32B and 72B, 80 GB GPU, int4	0.4 and 0.9	1.5 and 3	0.3 and 0.7

Chain-of-thought with 256 new tokens costs roughly ten times as much as direct generation, and thinking with up to 2048 tokens up to fifty times as much. LoRA training takes two to ten minutes per cell for a 7B model on one GPU and

under a minute for models below 1B. In memory, bfloat16 weights need about 2 bytes per parameter and 4-bit weights about 0.6, plus a small key-value cache for these short prompts.

Counted with `deadeye estimate`, which rolls out the oracle on the first ten seeds of each task configuration, one model and method in the main sweep make 5000 decisions on the bandit, 3000 on the contextual bandit, 2000 on loan approval, about 660 on gridworld, 330 on tic-tac-toe and 480 on blackjack, about 11500 in all. Weaker policies take more steps on gridworld, so these counts are lower bounds there. With 26 model entries and five language-model method configurations (the feature probe runs once, without a model), the main sweep makes about 1.5 million language-model decisions before probe and LoRA training; `deadeye estimate`, which multiplies by all six method configurations, reports about 1.8 million. The planning estimate in `RESEARCH_PLAN.md` is about 120 GPU-hours on one 80 GB GPU. The ablation configuration crosses 12 models, 16 method configurations and 12 task configurations, about 6.1 million decisions, roughly four times the main sweep and with the most expensive reasoning budgets. **[TODO: RESEARCH_PLAN.md and docs/compute_budget.md put the ablations at about the same cost as the sweep; trim the ablation cross product to the pre-registered contrasts or revise the budget.]** The CPU pilot takes six to twelve hours on a laptop.

13 Reproducibility checklist

The commands below rebuild every result in the paper from a fresh clone. Each step can be split across machines, since the runner skips cells whose summary already exists; `scripts/run_sweep.sh` assigns models to GPUs.

```
git clone https://github.com/CaptiveON/DeadEye && cd DeadEye
git checkout prereg-v1 # frozen prompts, configs and analysis code
python -m venv .venv && source .venv/bin/activate
pip install torch --index-url https://download.pytorch.org/whl/cpu # or a CUDA build
pip install -e ".[hf,dev]" tabulate # add ".[quant]" for int4/int8 on CUDA
pytest -q # unit and integration tests, no downloads
deadeye smoke # every task and method, mock and tiny random
model
deadeye show-prompt gridworld --seed 3 --cot --few-shot 2
deadeye estimate configs/sweep_gpu.yaml --seconds-per-decision 0.3
deadeye run configs/pilot_cpu.yaml # Phase 1 pilot
deadeye run configs/sweep_gpu.yaml # main sweep; split with --only-model
deadeye run configs/sweep_controls.yaml # Pythia and OLMo 2 controls
deadeye run configs/ablations/quantisation.yaml # one block at a time: quantisation,
# prompting, thinking, adaptation, tasks
deadeye report results/sweep_gpu --out report/sweep_gpu
deadeye compare results/sweep_gpu \
  --pair "loan:Qwen__Qwen2.5-1.5B-Instruct/prompt_score vs loan:Qwen__Qwen2.5-1.5B-
  Instruct/prompt_generate"
cd paper && make assets && make # regenerate tables and figures, build the PDF
```

- **Code and license.** The harness, configurations, prompts and analysis code are released under Apache-2.0; a Dockerfile builds a CPU image with the harness installed, whose entry point is the `deadeye` command.
- **Tests.** `pytest` checks that oracles return legal actions and episodes terminate, that a seed reproduces its instance, and that the oracle beats the random policy on every task. It checks the gridworld oracle against shortest paths; that the tic-tac-toe oracle never loses to the minimax opponent and that expectimax does at least as well as minimax against the random one; the blackjack table on known hands; and that the dealer's cards do not depend on the player's hits. On the model side it checks exact scoring against a manual log-probability computation, including labels that are token prefixes of one another, that greedy generation is deterministic and ignores checkpoint repetition penalties, that hidden states differ by layer, that LoRA training changes the model, and that base checkpoints are rendered without a chat template. It also covers the parser on realistic outputs, the prompt structure, the illegal-action policies and the rule that fallback actions never count as agreement, the rejection of

overlapping training and evaluation seeds, the selection of training variants, the runner end to end, and the report on smoke-test results.

- **Determinism.** Instances, reward tables and card streams are fixed by the seed; greedy decoding, scoring and probes reproduce exactly on the same hardware and library versions.
- **Provenance.** Every cell’s `summary.json` records the model and method specification, precision, parameter count, seeds and timings, and `run_manifest.json` the configuration and library versions (Section 3.6). **[TODO: Hardware and library versions as in Section 4.4.]**
- **Released data.** `summary.json` and `episodes.jsonl` for every cell, the per-decision `steps.jsonl` as an archive, and the generated report. **[TODO: Archive DOI.]**
- **Exclusions and deviations.** Listed in Appendix C.

14 Model ladders: verified repository identifiers

Table 11 lists the Hugging Face repository identifiers that `docs/model_ladders.md` records as verified, for the six ladders we use; an identifier counts as verified when it was seen exactly in an official listing. Cells marked “not verified” are configured checkpoints whose identifiers follow the family’s naming pattern but have not been confirmed. Qwen3 also has separately trained instruct and thinking checkpoints at 4B, Qwen/Qwen3-4B-Instruct-2507 and Qwen/Qwen3-4B-Thinking-2507, both verified, which offer a second contrast for H7. **[TODO: Verify the remaining identifiers before the sweep: Qwen2.5 1.5B and 7B base, Qwen2.5 14B and 32B Instruct, the Qwen3 0.6B, 1.7B, 4B and 14B post-trained checkpoints, SmolLM2 1.7B Instruct and the SmolLM2 360M base used in the pilot, and the Gemma 3 270M and 12B instruction-tuned checkpoints. `configs/models.yaml` marks several of these as verified, which disagrees with `docs/model_ladders.md`.]**

Verified repository identifiers. For Pythia, the non-deduplicated identifiers (without `-deduped`) are verified as well; Pythia has no instruct checkpoints, and Qwen3 has no 32B base checkpoint.

Family	Size	Base or pre-trained	Instruct or post-trained
Qwen2.5	0.5B	Qwen/Qwen2.5-0.5B	Qwen/Qwen2.5-0.5B-Instruct
	1.5B	not verified	Qwen/Qwen2.5-1.5B-Instruct
	3B	Qwen/Qwen2.5-3B	Qwen/Qwen2.5-3B-Instruct
	7B	not verified	Qwen/Qwen2.5-7B-Instruct
	14B	Qwen/Qwen2.5-14B	not verified
	32B	Qwen/Qwen2.5-32B	not verified
	72B	Qwen/Qwen2.5-72B	Qwen/Qwen2.5-72B-Instruct
Qwen3	0.6B, 1.7B, 4B	not verified	not verified
	8B	Qwen/Qwen3-8B-Base	Qwen/Qwen3-8B
	14B	Qwen/Qwen3-14B-Base	not verified
	32B	none released	Qwen/Qwen3-32B
SmolLM2	135M	not verified	HuggingFaceTB/SmolLM2-135M-Instruct
	360M	not verified	HuggingFaceTB/SmolLM2-360M-Instruct
	1.7B	HuggingFaceTB/SmolLM2-1.7B	not verified
Gemma 3	270M, 12B	not verified	not verified
	1B	google/gemma-3-1b-pt	google/gemma-3-1b-it
	4B	google/gemma-3-4b-pt	google/gemma-3-4b-it

Family	Size	Base or pre-trained	Instruct or post-trained
	27B	not verified	google/gemma-3-27b-it
Pythia	70M	EleutherAI/pythia-70m-deduped	
	160M	EleutherAI/pythia-160m-deduped	
	410M	EleutherAI/pythia-410m-deduped	
	1B	EleutherAI/pythia-1b-deduped	
	1.4B	EleutherAI/pythia-1.4b-deduped	
	2.8B	EleutherAI/pythia-2.8b-deduped	
	6.9B	EleutherAI/pythia-6.9b-deduped	
	12B	EleutherAI/pythia-12b-deduped	
OLMo 2	1B	allenai/OLMo-2-0425-1B	allenai/OLMo-2-0425-1B-Instruct
	7B	allenai/OLMo-2-1124-7B	allenai/OLMo-2-1124-7B-Instruct
	13B	allenai/OLMo-2-1124-13B	allenai/OLMo-2-1124-13B-Instruct
	32B	allenai/OLMo-2-0325-32B	allenai/OLMo-2-0325-32B-Instruct

References

- Agarwal, R. *et al.* (2021) “Deep reinforcement learning at the edge of the statistical precipice,” *Advances in neural information processing systems 34 (NeurIPS 2021)*. Curran Associates, Inc. Available at: <https://arxiv.org/abs/2108.13264>.
- Auer, P., Cesa-Bianchi, N. and Fischer, P. (2002) “Finite-time analysis of the multiarmed bandit problem,” *Machine Learning*, 47, pp. 235–256. Available at: <https://doi.org/10.1023/A:1013689704352>.
- Baldwin, R.R. *et al.* (1956) “The optimum strategy in blackjack,” *Journal of the American Statistical Association*, 51(275), pp. 429–439. Available at: <https://doi.org/10.1080/01621459.1956.10501334>.
- Belinkov, Y. (2022) “Probing classifiers: Promises, shortcomings, and advances,” *Computational Linguistics*, 48(1), pp. 207–219. Available at: https://doi.org/10.1162/coli_a_00422.
- Ben Allal, L. *et al.* (2025) “SmolLM2: When smol goes big – data-centric training of a small language model,” *arXiv preprint arXiv:2502.02737* [Preprint]. Available at: <https://arxiv.org/abs/2502.02737>.
- Biderman, S. *et al.* (2023) “Pythia: A suite for analyzing large language models across training and scaling,” *Proceedings of the 40th international conference on machine learning*. PMLR (Proceedings of machine learning research), pp. 2397–2430. Available at: <https://arxiv.org/abs/2304.01373>.
- Binz, M. *et al.* (2025) “A foundation model to predict and capture human cognition,” *Nature*, 644(8078), pp. 1002–1009. Available at: <https://doi.org/10.1038/s41586-025-09215-4>.
- Binz, M. and Schulz, E. (2023) “Using cognitive psychology to understand GPT-3,” *Proceedings of the National Academy of Sciences*, 120(6), p. e2218523120. Available at: <https://doi.org/10.1073/pnas.2218523120>.
- Brown, T.B. *et al.* (2020) “Language models are few-shot learners,” *Advances in neural information processing systems 33 (NeurIPS 2020)*. Curran Associates, Inc. Available at: <https://arxiv.org/abs/2005.14165>.
- Carta, T. *et al.* (2023) “Grounding large language models in interactive environments with online reinforcement learning,” *Proceedings of the 40th international conference on machine learning*. PMLR (Proceedings of machine learning research), pp. 3676–3713. Available at: <https://arxiv.org/abs/2302.02662>.
- Chu, T. *et al.* (2025) “SFT memorizes, RL generalizes: A comparative study of foundation model post-training,” *Proceedings of the 42nd international conference on machine learning*. PMLR (Proceedings of machine learning research), pp. 10818–10838. Available at: <https://arxiv.org/abs/2501.17161>.
- Cloudflare (2026) “Introducing Clef: Our open-source decision models, and new RL fine-tuning platform.” Cloudflare blog, 1 October 2026, <https://blog.cloudflare.com/clef-decision-models/>.
- Convai Innovations (2026) “Laya: Open System One decision models (model cards for laya, laya-multilingual and laya-typed-decisions).” <https://huggingface.co/convaiinnovations/laya>; <https://huggingface.co/convaiinnovations/laya-typed-decisions>; <https://laya.convaiinnovations.com/>.

- Dettmers, T. *et al.* (2022) “GPT3.int8(): 8-bit matrix multiplication for transformers at scale,” *Advances in neural information processing systems* 35 (*NeurIPS* 2022). Curran Associates, Inc. Available at: <https://arxiv.org/abs/2208.07339>.
- Dettmers, T. *et al.* (2023) “QLoRA: Efficient finetuning of quantized LLMs,” *Advances in neural information processing systems* 36 (*NeurIPS* 2023). Curran Associates, Inc. Available at: <https://arxiv.org/abs/2305.14314>.
- Dettmers, T. and Zettlemoyer, L. (2023) “The case for 4-bit precision: K-bit inference scaling laws,” *Proceedings of the 40th international conference on machine learning*. PMLR (Proceedings of machine learning research), pp. 7750–7774. Available at: <https://arxiv.org/abs/2212.09720>.
- Duan, J. *et al.* (2024) “GTBench: Uncovering the strategic reasoning capabilities of LLMs via game-theoretic evaluations,” *Advances in neural information processing systems* 37 (*NeurIPS* 2024). Curran Associates, Inc. Available at: <https://arxiv.org/abs/2402.12348>.
- Efron, B. and Tibshirani, R.J. (1993) *An introduction to the bootstrap*. New York: Chapman & Hall (Monographs on statistics and applied probability, 57). Available at: <https://doi.org/10.1201/9780429246593>.
- Fan, C. *et al.* (2024) “Can large language models serve as rational players in game theory? A systematic analysis,” *Proceedings of the AAAI Conference on Artificial Intelligence*, 38(16), pp. 17960–17967. Available at: <https://doi.org/10.1609/aaai.v38i16.29751>.
- Frantar, E. *et al.* (2023) “OPTQ: Accurate quantization for generative pre-trained transformers,” *The eleventh international conference on learning representations (ICLR 2023)*. OpenReview.net. Available at: <https://arxiv.org/abs/2210.17323>.
- Gema, A.P. *et al.* (2025) “Inverse scaling in test-time compute,” *Transactions on Machine Learning Research* [Preprint]. Available at: <https://openreview.net/forum?id=NXgyHW1c7M>.
- Gemma Team (2025) “Gemma 3 technical report,” *arXiv preprint arXiv:2503.19786* [Preprint]. Available at: <https://arxiv.org/abs/2503.19786>.
- Grattafiori, A., Dubey, A., *et al.* (2024) “The Llama 3 herd of models,” *arXiv preprint arXiv:2407.21783* [Preprint]. Available at: <https://arxiv.org/abs/2407.21783>.
- Guo, D. *et al.* (2025) “DeepSeek-R1 incentivizes reasoning in LLMs through reinforcement learning,” *Nature*, 645(8081), pp. 633–638. Available at: <https://doi.org/10.1038/s41586-025-09422-z>.
- Harris, K. and Slivkins, A. (2026) “Should you use your large language model to explore or exploit?” *Proceedings of the 42nd conference on uncertainty in artificial intelligence*. PMLR (Proceedings of machine learning research), pp. 2008–2058. Available at: <https://arxiv.org/abs/2502.00225>.
- Hegselmann, S. *et al.* (2023) “TabLLM: Few-shot classification of tabular data with large language models,” *Proceedings of the 26th international conference on artificial intelligence and statistics*. PMLR (Proceedings of machine learning research), pp. 5549–5581. Available at: <https://arxiv.org/abs/2210.10723>.
- Henderson, P. *et al.* (2018) “Deep reinforcement learning that matters,” *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1). Available at: <https://doi.org/10.1609/aaai.v32i1.11694>.
- Hoffmann, J. *et al.* (2022) “An empirical analysis of compute-optimal large language model training,” *Advances in neural information processing systems* 35 (*NeurIPS* 2022). Curran Associates, Inc. Available at: <https://arxiv.org/abs/2203.15556>.
- Holm, S. (1979) “A simple sequentially rejective multiple test procedure,” *Scandinavian Journal of Statistics*, 6(2), pp. 65–70.
- Holtzman, A. *et al.* (2021) “Surface form competition: Why the highest probability answer isn’t always right,” *Proceedings of the 2021 conference on empirical methods in natural language processing*. Online; Punta Cana, Dominican Republic: Association for Computational Linguistics, pp. 7038–7051. Available at: <https://doi.org/10.18653/v1/2021.emnlp-main.564>.
- Hu, E.J. *et al.* (2022) “LoRA: Low-rank adaptation of large language models,” *The tenth international conference on learning representations (ICLR 2022)*. OpenReview.net. Available at: <https://arxiv.org/abs/2106.09685>.
- Huang, W. *et al.* (2022) “Language models as zero-shot planners: Extracting actionable knowledge for embodied agents,” *Proceedings of the 39th international conference on machine learning*. PMLR (Proceedings of machine learning research), pp. 9118–9147. Available at: <https://arxiv.org/abs/2201.07207>.
- Jia, J. *et al.* (2024) “Decision-making behavior evaluation framework for LLMs under uncertain context,” *Advances in neural information processing systems* 37 (*NeurIPS* 2024). Curran Associates, Inc. Available at: <https://arxiv.org/abs/2406.05972>.
- Jiawode, M. *et al.* (2026) “Spatial reasoning in LLM game agents: Impact of causal context and multi-step planning,” *arXiv preprint arXiv:2607.22732* [Preprint]. Available at: <https://arxiv.org/abs/2607.22732>.
- Krishnamurthy, A. *et al.* (2024) “Can large language models explore in-context?” *Advances in neural information processing systems* 37 (*NeurIPS* 2024). Curran Associates, Inc. Available at: <https://arxiv.org/abs/2403.15371>.
- Kumar, T. *et al.* (2025) “Scaling laws for precision,” *The thirteenth international conference on learning representations (ICLR 2025)*. OpenReview.net. Available at: <https://arxiv.org/abs/2411.04330>.

- Laskin, M. *et al.* (2023) “In-context reinforcement learning with algorithm distillation,” *The eleventh international conference on learning representations (ICLR 2023)*. OpenReview.net. Available at: <https://arxiv.org/abs/2210.14215>.
- Lattimore, T. and Szepesvári, C. (2020) *Bandit algorithms*. Cambridge: Cambridge University Press. Available at: <https://doi.org/10.1017/9781108571401>.
- Law, A.M. (2015) *Simulation modeling and analysis*. Fifth. New York: McGraw-Hill Education.
- Lee, T. (2026) “SemIf (formerly OpenJev): Semantic ifs from open models.” Software, <https://github.com/TheoLeeCJ/SemIf-OpenJev>.
- Li, Y. *et al.* (2025) “Small models struggle to learn from strong reasoners,” *Findings of the association for computational linguistics: ACL 2025*. Vienna, Austria: Association for Computational Linguistics, pp. 25366–25394. Available at: <https://doi.org/10.18653/v1/2025.findings-acl.1301>.
- Lin, B.Y. *et al.* (2024) “The unlocking spell on base LLMs: Rethinking alignment via in-context learning,” *The twelfth international conference on learning representations (ICLR 2024)*. OpenReview.net. Available at: <https://arxiv.org/abs/2312.01552>.
- Lin, J. *et al.* (2024) “AWQ: Activation-aware weight quantization for on-device LLM compression and acceleration,” *Proceedings of machine learning and systems 6 (MLSys 2024)*. Available at: <https://arxiv.org/abs/2306.00978>.
- Liu, B. *et al.* (2026) “SPIRAL: Self-play on zero-sum games incentivizes reasoning via multi-agent multi-turn reinforcement learning,” *The fourteenth international conference on learning representations (ICLR 2026)*. OpenReview.net. Available at: <https://arxiv.org/abs/2506.24119>.
- Liu, R., Geng, J., Peterson, J.C., *et al.* (2025) “Large language models assume people are more rational than we really are,” *The thirteenth international conference on learning representations (ICLR 2025)*. OpenReview.net. Available at: <https://arxiv.org/abs/2406.17055>.
- Liu, R., Geng, J., Wu, A.J., *et al.* (2025) “Mind your step (by step): Chain-of-thought can reduce performance on tasks where thinking makes humans worse,” *Proceedings of the 42nd international conference on machine learning*. PMLR (Proceedings of machine learning research), pp. 38489–38517. Available at: <https://arxiv.org/abs/2410.21333>.
- Liu, X. *et al.* (2024) “AgentBench: Evaluating LLMs as agents,” *The twelfth international conference on learning representations (ICLR 2024)*. OpenReview.net. Available at: <https://arxiv.org/abs/2308.03688>.
- Monea, G. *et al.* (2025) “LLMs are in-context bandit reinforcement learners,” *Second conference on language modeling (COLM 2025)*. OpenReview.net. Available at: <https://arxiv.org/abs/2410.05362>.
- Nie, A. *et al.* (2025) “EVOlVE: Evaluating and optimizing LLMs for in-context exploration,” *Proceedings of the 42nd international conference on machine learning*. PMLR (Proceedings of machine learning research), pp. 46346–46376. Available at: <https://arxiv.org/abs/2410.06238>.
- Orgad, H. *et al.* (2025) “LLMs know more than they show: On the intrinsic representation of LLM hallucinations,” *The thirteenth international conference on learning representations (ICLR 2025)*. OpenReview.net. Available at: <https://arxiv.org/abs/2410.02707>.
- Paglieri, D. *et al.* (2025) “BALROG: Benchmarking agentic LLM and VLM reasoning on games,” *The thirteenth international conference on learning representations (ICLR 2025)*. OpenReview.net. Available at: <https://arxiv.org/abs/2411.13543>.
- Palmer, J. (2026) “Kev: Open decision models on Qwen3.5 and Qwen3.8 with a System One-compatible server.” Software and model cards, <https://github.com/jaredpalmer/kev>; <https://huggingface.co/jaredpalmer/kev-9b>.
- Pedregosa, F. *et al.* (2011) “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, 12(85), pp. 2825–2830.
- Perplexity AI (2026) “Pplx-decider-v1.1-27b: Model card.” <https://huggingface.co/perplexity-ai/pplx-decider-v1.1-27b>.
- Pomerleau, D.A. (1988) “ALVINN: An autonomous land vehicle in a neural network,” in D.S. Touretzky (ed.) *Advances in neural information processing systems 1 (NIPS 1988)*. Morgan Kaufmann, pp. 305–313.
- Rafailov, R. *et al.* (2023) “Direct preference optimization: Your language model is secretly a reward model,” *Advances in neural information processing systems 36 (NeurIPS 2023)*. Curran Associates, Inc. Available at: <https://arxiv.org/abs/2305.18290>.
- Robinson, J. and Wingate, D. (2023) “Leveraging large language models for multiple choice question answering,” *The eleventh international conference on learning representations (ICLR 2023)*. OpenReview.net. Available at: <https://arxiv.org/abs/2210.12353>.
- Ross, S., Gordon, G. and Bagnell, D. (2011) “A reduction of imitation learning and structured prediction to no-regret online learning,” *Proceedings of the fourteenth international conference on artificial intelligence and statistics*. Fort Lauderdale, FL, USA: PMLR (Proceedings of machine learning research), pp. 627–635. Available at: <https://arxiv.org/abs/1011.0686>.
- Ruoss, A. *et al.* (2025) “LMAct: A benchmark for in-context imitation learning with long multimodal demonstrations,” *Proceedings of the 42nd international conference on machine learning*. PMLR (Proceedings of machine learning research), pp. 52303–52344. Available at: <https://arxiv.org/abs/2412.01441>.
- Schaeffer, R., Miranda, B. and Koyejo, S. (2023) “Are emergent abilities of large language models a mirage?” *Advances in neural information processing systems 36 (NeurIPS 2023)*. Curran Associates, Inc. Available at: <https://arxiv.org/abs/2304.15004>.

- Schmied, T. *et al.* (2026) “LLMs are greedy agents: Effects of RL fine-tuning on decision-making abilities,” *The fourteenth international conference on learning representations (ICLR 2026)*. OpenReview.net. Available at: <https://arxiv.org/abs/2504.16078>.
- Sciar, M. *et al.* (2024) “Quantifying language models’ sensitivity to spurious features in prompt design or: How I learned to start worrying about prompt formatting,” *The twelfth international conference on learning representations (ICLR 2024)*. OpenReview.net. Available at: <https://arxiv.org/abs/2310.11324>.
- Shao, Z. *et al.* (2024) “DeepSeekMath: Pushing the limits of mathematical reasoning in open language models,” *arXiv preprint arXiv:2402.03300* [Preprint]. Available at: <https://arxiv.org/abs/2402.03300>.
- Shojaee, P. *et al.* (2025) “The illusion of thinking: Understanding the strengths and limitations of reasoning models via the lens of problem complexity,” *Advances in neural information processing systems 38 (NeurIPS 2025)*. Curran Associates, Inc. Available at: <https://arxiv.org/abs/2506.06941>.
- Sinha, A. *et al.* (2026) “The illusion of diminishing returns: Measuring long horizon execution in LLMs,” *The fourteenth international conference on learning representations (ICLR 2026)*. OpenReview.net. Available at: <https://arxiv.org/abs/2509.09677>.
- Skean, O. *et al.* (2025) “Layer by layer: Uncovering hidden representations in language models,” *Proceedings of the 42nd international conference on machine learning*. PMLR (Proceedings of machine learning research), pp. 55854–55875. Available at: <https://arxiv.org/abs/2502.02013>.
- Snell, C. *et al.* (2025) “Scaling LLM test-time compute optimally can be more effective than scaling parameters for reasoning,” *The thirteenth international conference on learning representations (ICLR 2025)*. OpenReview.net. Available at: <https://arxiv.org/abs/2408.03314>.
- Sutton, R.S. and Barto, A.G. (2018) *Reinforcement learning: An introduction*. Second. Cambridge, MA: MIT Press (Adaptive computation and machine learning).
- Szot, A. *et al.* (2024) “Large language models as generalizable policies for embodied tasks,” *The twelfth international conference on learning representations (ICLR 2024)*. OpenReview.net. Available at: <https://arxiv.org/abs/2310.17722>.
- Tam, Z.R. *et al.* (2024) “Let me speak freely? A study on the impact of format restrictions on large language model performance,” *Proceedings of the 2024 conference on empirical methods in natural language processing: Industry track*. Miami, Florida, US: Association for Computational Linguistics, pp. 1218–1236. Available at: <https://doi.org/10.18653/v1/2024.emnlp-industry.91>.
- Tan, W. *et al.* (2024) “True knowledge comes from practice: Aligning large language models with embodied environments via reinforcement learning,” *The twelfth international conference on learning representations (ICLR 2024)*. OpenReview.net. Available at: <https://arxiv.org/abs/2401.14151>.
- Team Olmo, Ettinger, A., *et al.* (2025) “Olmo 3,” *arXiv preprint arXiv:2512.13961* [Preprint]. Available at: <https://arxiv.org/abs/2512.13961>.
- TechCrunch (2026a) “A new kind of AI model from a ChatGPT inventor is thrilling developers.” TechCrunch, 18 September 2026, <https://techcrunch.com/2026/09/18/a-new-kind-of-ai-model-from-a-chatgpt-inventor-is-thrilling-developers/>.
- TechCrunch (2026b) “Amazon releases its own Jev clone as decision models flood the web.” TechCrunch, 1 October 2026, <https://techcrunch.com/2026/10/01/amazon-releases-its-own-jev-clone-as-decision-models-flood-the-web/>.
- The New Stack (2026) “TypeSafe launched Jev because sequential LLMs are ‘totally useless for computers’.” <https://thenewstack.io/typesafe-jev-system-one/>.
- Towers, M. *et al.* (2025) “Gymnasium: A standard interface for reinforcement learning environments,” *Advances in neural information processing systems 38 (NeurIPS 2025), datasets and benchmarks track*. Curran Associates, Inc. Available at: <https://doi.org/10.52202/085713-4916>.
- TypeSafe AI (2026a) “Choice: TypeSafe AI documentation.” <https://docs.typesafe.ai/primitives/choice>.
- TypeSafe AI (2026b) “Introducing System One models & Jev.” TypeSafe AI blog, <https://typesafe.ai/blog/introducing-system-one-models-and-jev>.
- TypeSafe AI (2026c) “System One: TypeSafe AI documentation.” <https://docs.typesafe.ai/concepts/system-one>.
- Walsh, E.P. *et al.* (2025) “2 OLMo 2 Furious (COLM’s version),” *Second conference on language modeling (COLM 2025)*. OpenReview.net. Available at: <https://arxiv.org/abs/2501.00656>.
- Wang, X. *et al.* (2024) “‘My answer is C’: First-token probabilities do not match text answers in instruction-tuned language models,” *Findings of the association for computational linguistics: ACL 2024*. Bangkok, Thailand: Association for Computational Linguistics, pp. 7407–7416. Available at: <https://doi.org/10.18653/v1/2024.findings-acl.441>.
- Wei, J., Wang, X., *et al.* (2022) “Chain-of-thought prompting elicits reasoning in large language models,” *Advances in neural information processing systems 35 (NeurIPS 2022)*. Curran Associates, Inc. Available at: <https://doi.org/10.52202/068431-1800>.
- Wei, J., Tay, Y., *et al.* (2022) “Emergent abilities of large language models,” *Transactions on Machine Learning Research* [Preprint]. Available at: <https://openreview.net/forum?id=yzkSU5zdwD>.

- Wolf, T. *et al.* (2020) “Transformers: State-of-the-art natural language processing,” *Proceedings of the 2020 conference on empirical methods in natural language processing: System demonstrations*. Online: Association for Computational Linguistics, pp. 38–45. Available at: <https://doi.org/10.18653/v1/2020.emnlp-demos.6>.
- Wu, Y. *et al.* (2024) “SmartPlay: A benchmark for LLMs as intelligent agents,” *The twelfth international conference on learning representations (ICLR 2024)*. OpenReview.net. Available at: <https://arxiv.org/abs/2310.01557>.
- Yang, A. *et al.* (2024) “Qwen2.5 technical report,” *arXiv preprint arXiv:2412.15115* [Preprint]. Available at: <https://arxiv.org/abs/2412.15115>.
- Yang, A. *et al.* (2025) “Qwen3 technical report,” *arXiv preprint arXiv:2505.09388* [Preprint]. Available at: <https://arxiv.org/abs/2505.09388>.
- Yao, S. *et al.* (2023) “ReAct: Synergizing reasoning and acting in language models,” *The eleventh international conference on learning representations (ICLR 2023)*. OpenReview.net. Available at: <https://arxiv.org/abs/2210.03629>.
- Zhao, T.Z. *et al.* (2021) “Calibrate before use: Improving few-shot performance of language models,” *Proceedings of the 38th international conference on machine learning*. PMLR (Proceedings of machine learning research), pp. 12697–12706. Available at: <https://arxiv.org/abs/2102.09690>.
- Zheng, C. *et al.* (2024) “Large language models are not robust multiple choice selectors,” *The twelfth international conference on learning representations (ICLR 2024)*. OpenReview.net. Available at: <https://arxiv.org/abs/2309.03882>.
- Zhou, C. *et al.* (2023) “LIMA: Less is more for alignment,” *Advances in neural information processing systems 36 (NeurIPS 2023)*. Curran Associates, Inc. Available at: <https://arxiv.org/abs/2305.11206>.